

BFS Traversal

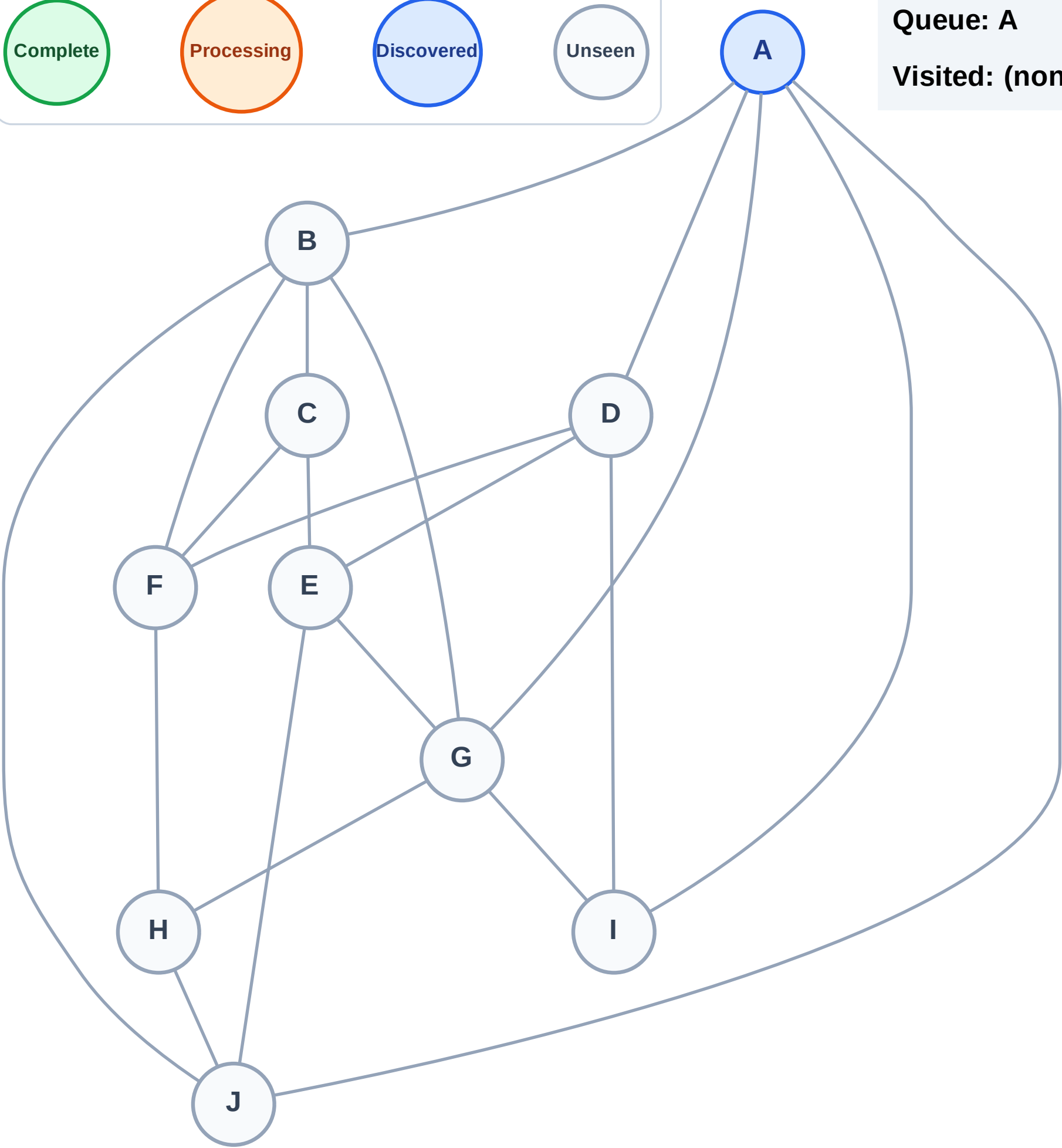
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

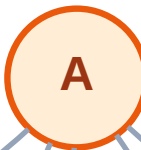
Step 2: Process node A

Legend



Queue: (empty)

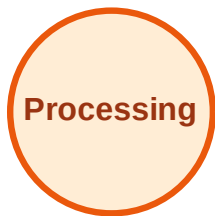
Visited: (none)



BFS Traversal

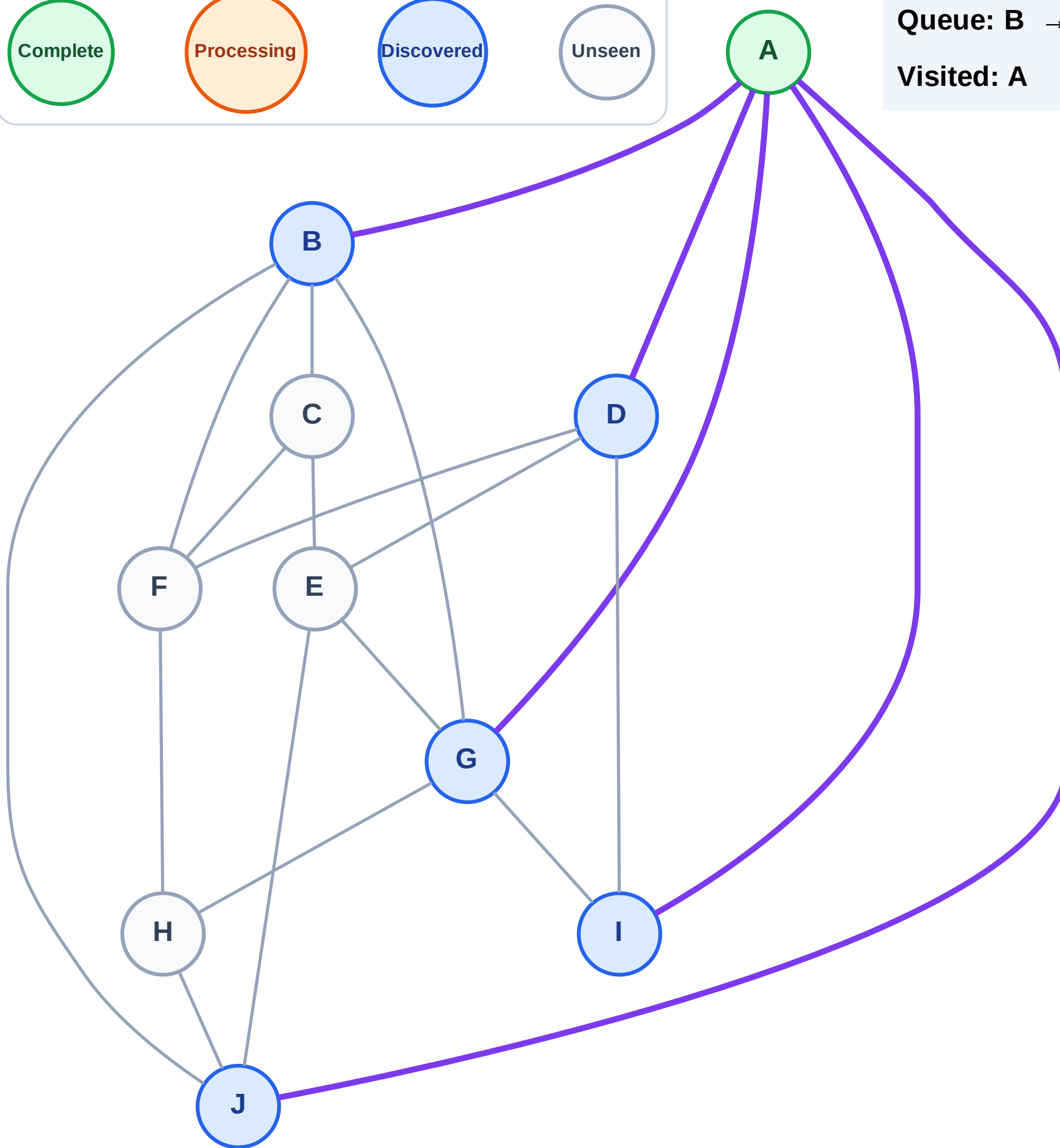
Step 3: Discover B, D, G, I, J from A

Legend



Queue: B → D → G → I → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

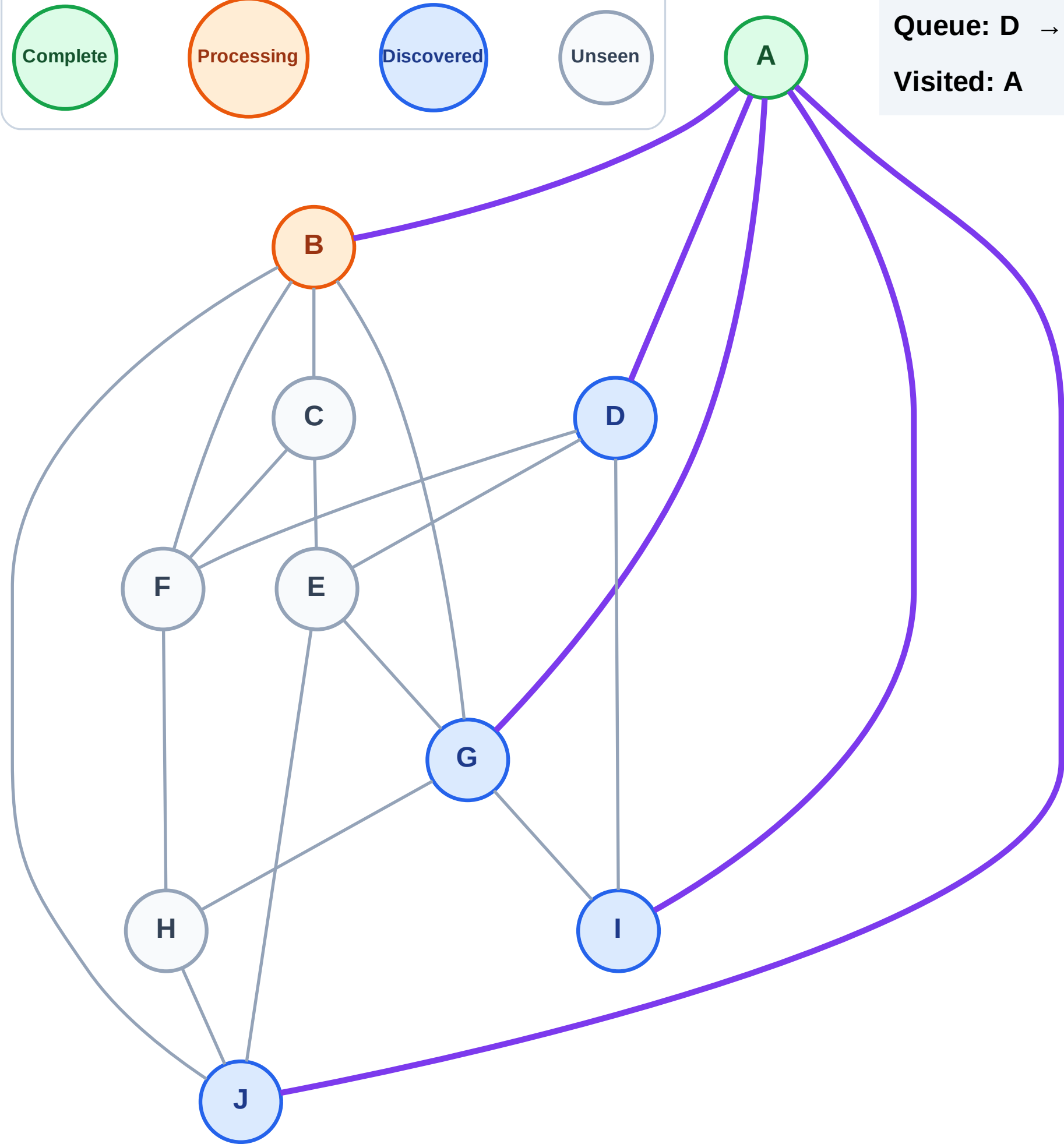
Processing

Discovered

Unseen

Queue: D → G → I → J

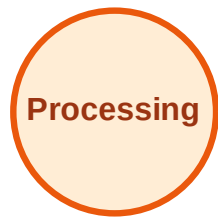
Visited: A



BFS Traversal

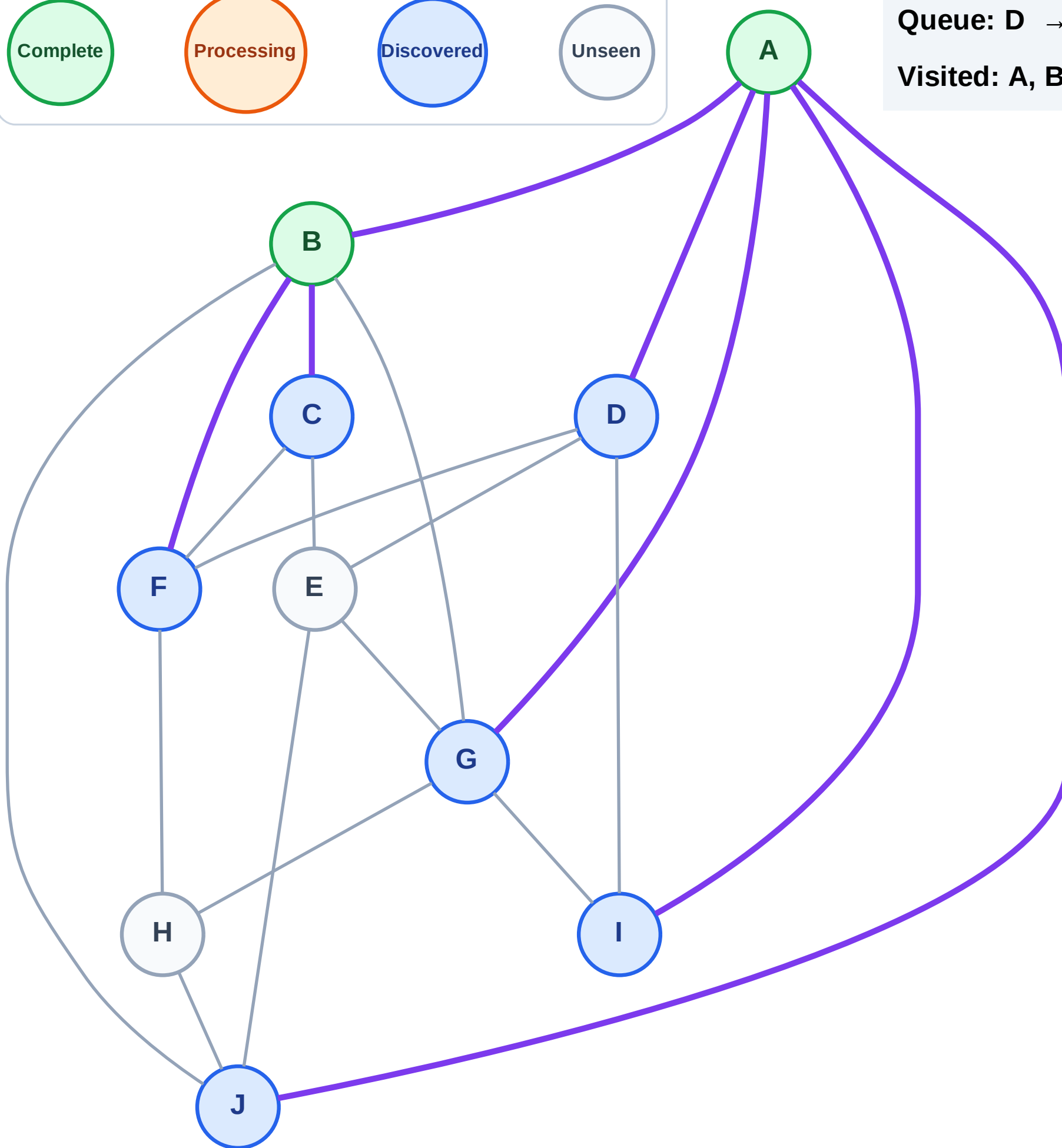
Step 5: Discover C, F from B

Legend



Queue: D → G → I → J → C → F

Visited: A, B



BFS Traversal

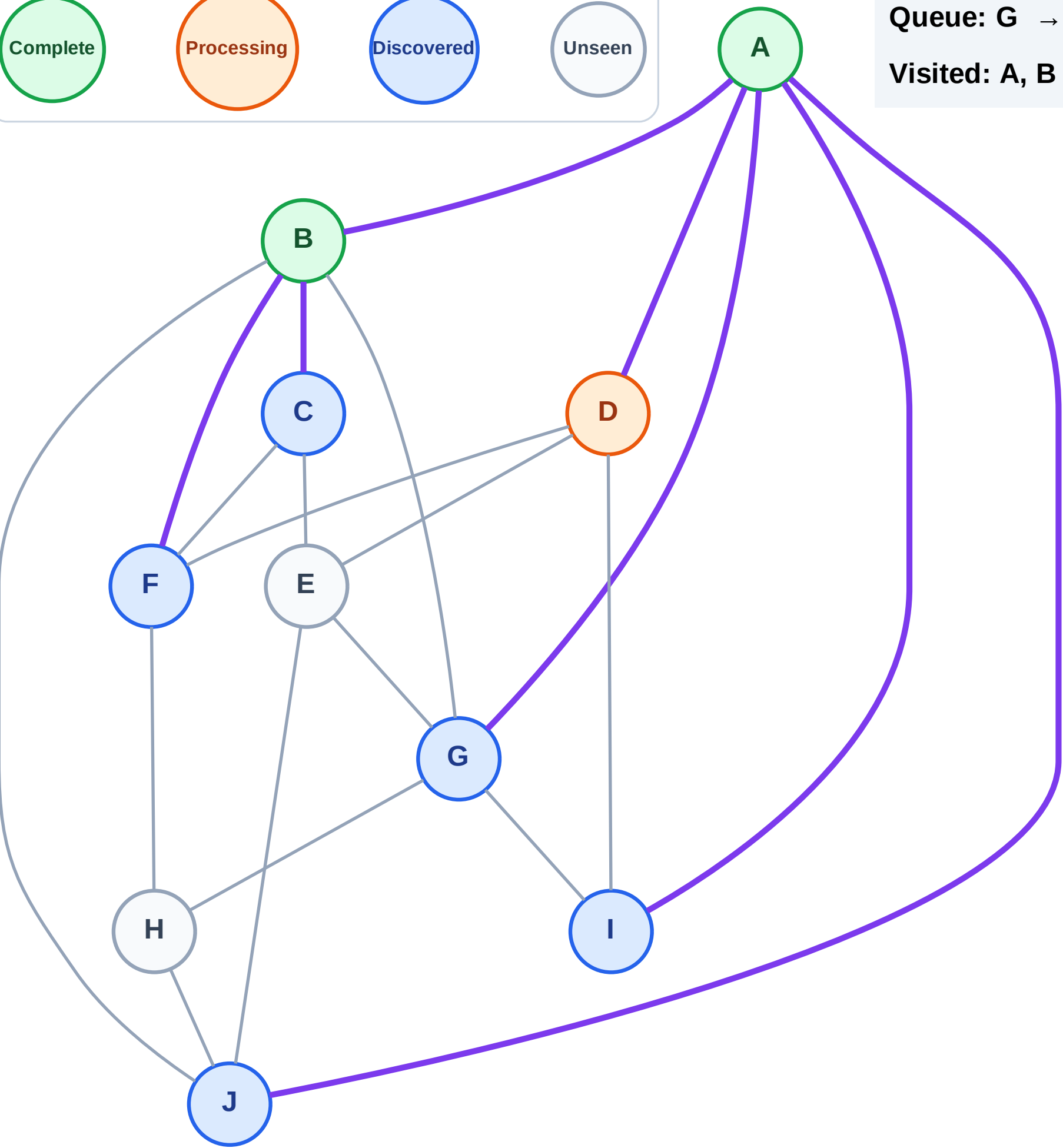
Step 6: Process node D

Legend



Queue: G → I → J → C → F

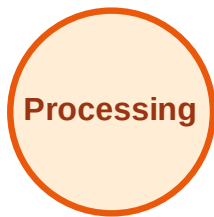
Visited: A, B



BFS Traversal

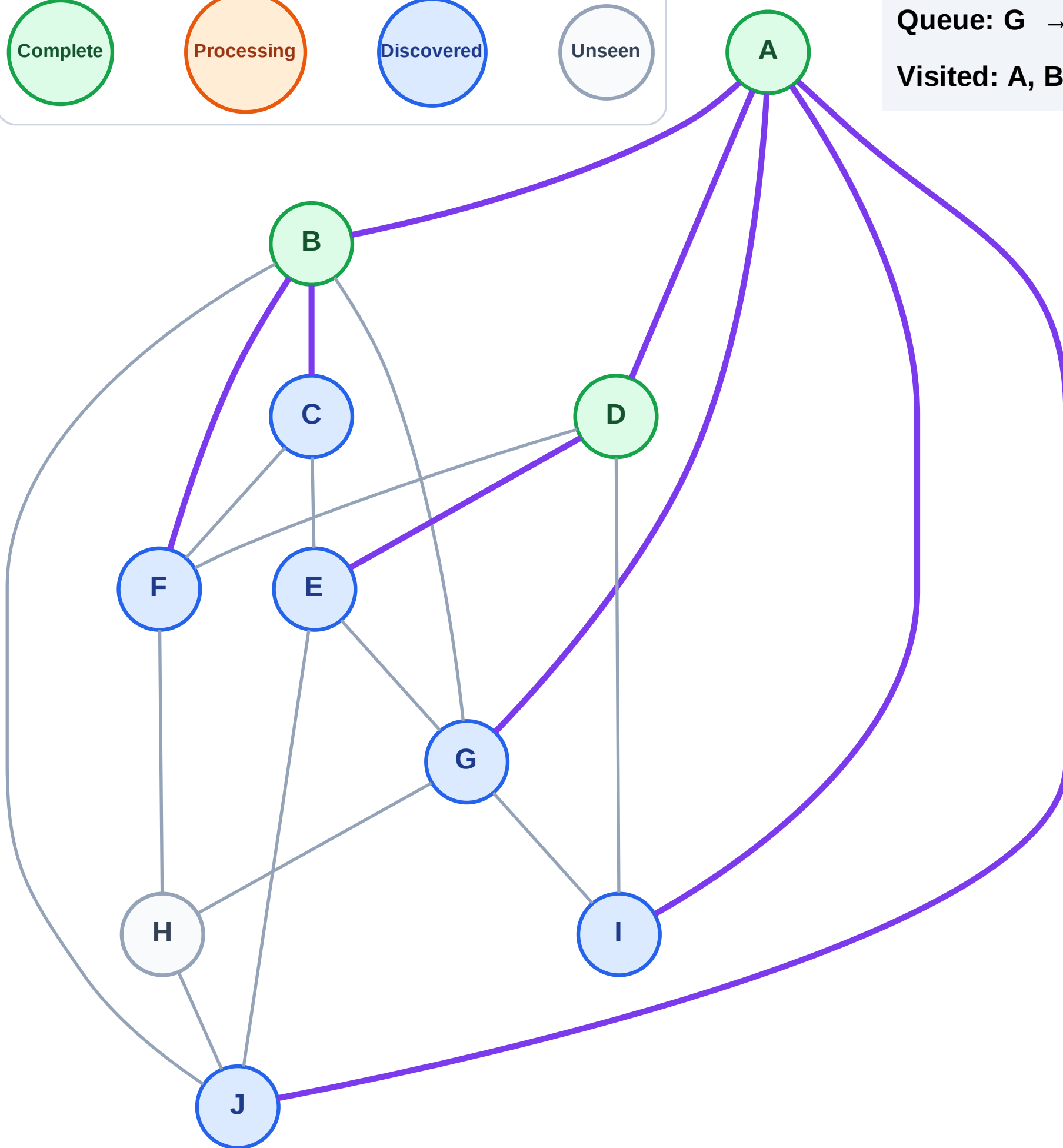
Step 7: Discover E from D

Legend

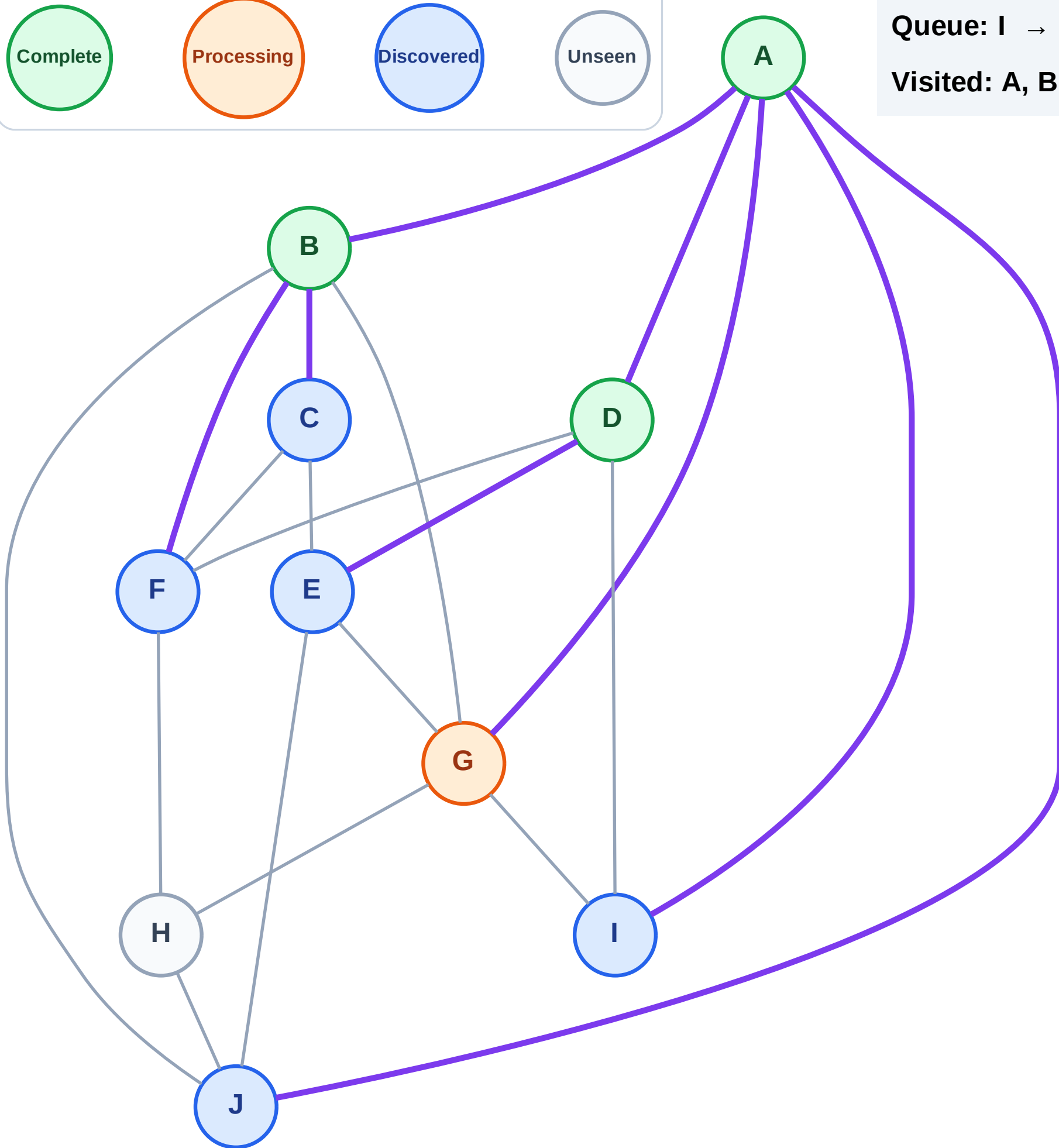
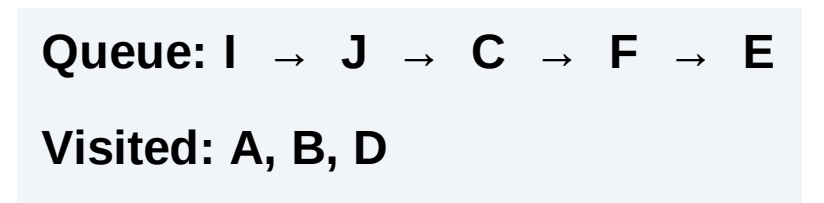


Queue: G → I → J → C → F → E

Visited: A, B, D

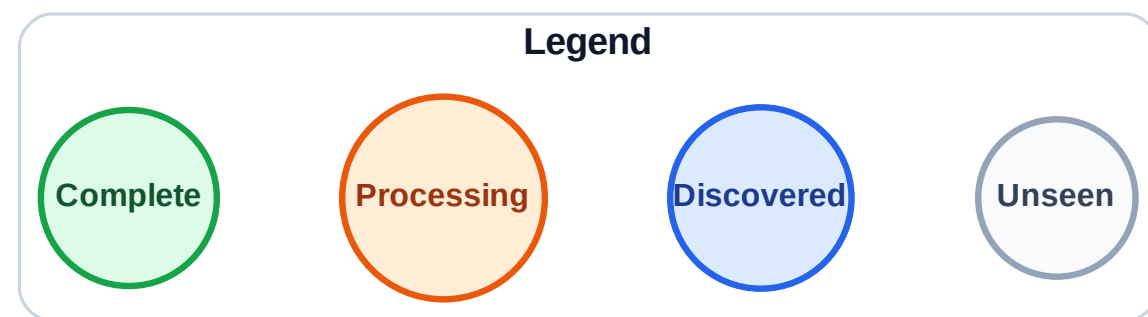


Step 8: Process node G



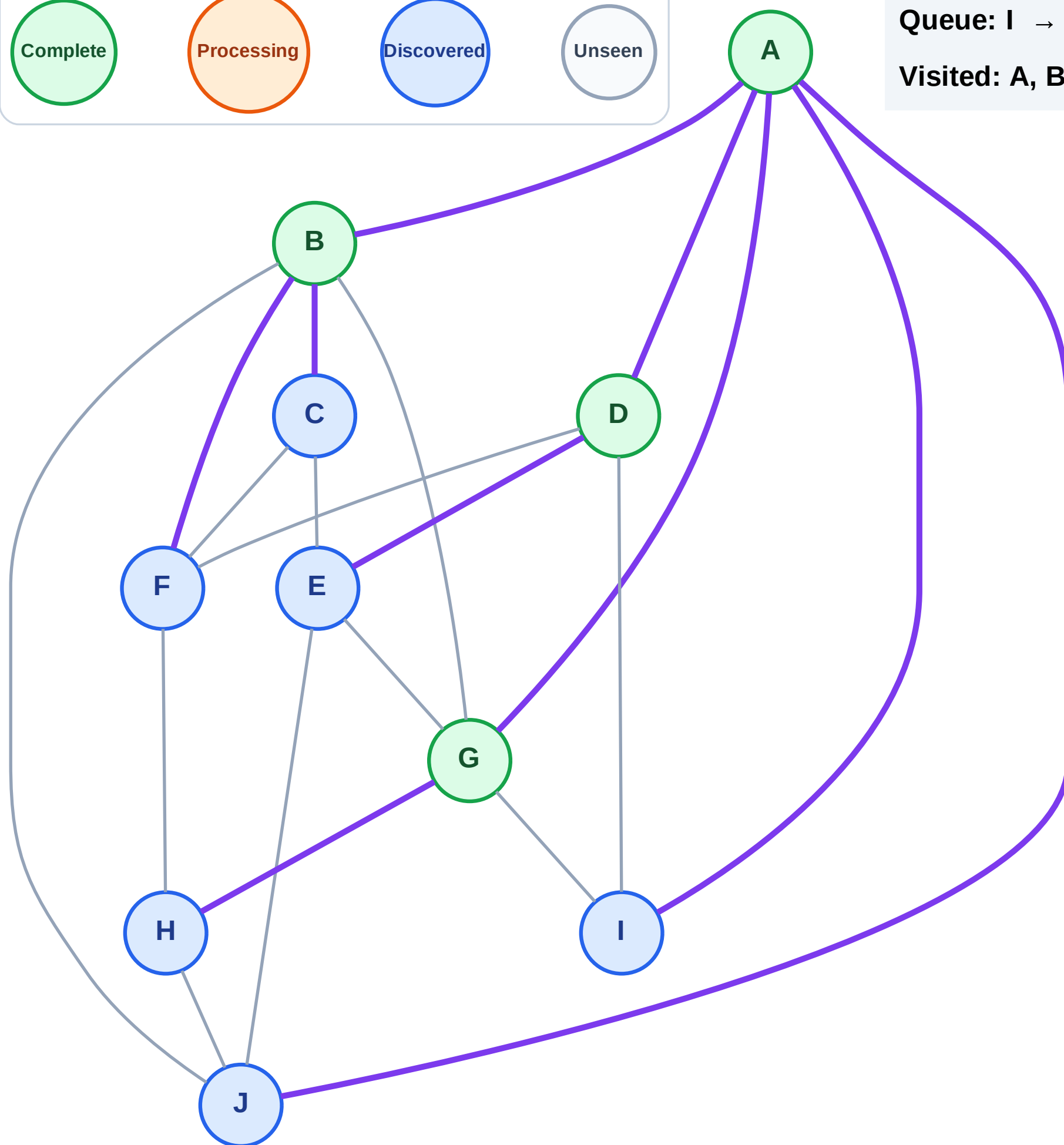
BFS Traversal

Step 9: Discover H from G



Queue: I → J → C → F → E → H

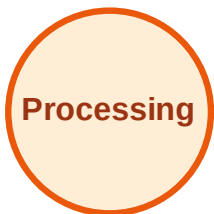
Visited: A, B, D, G



BFS Traversal

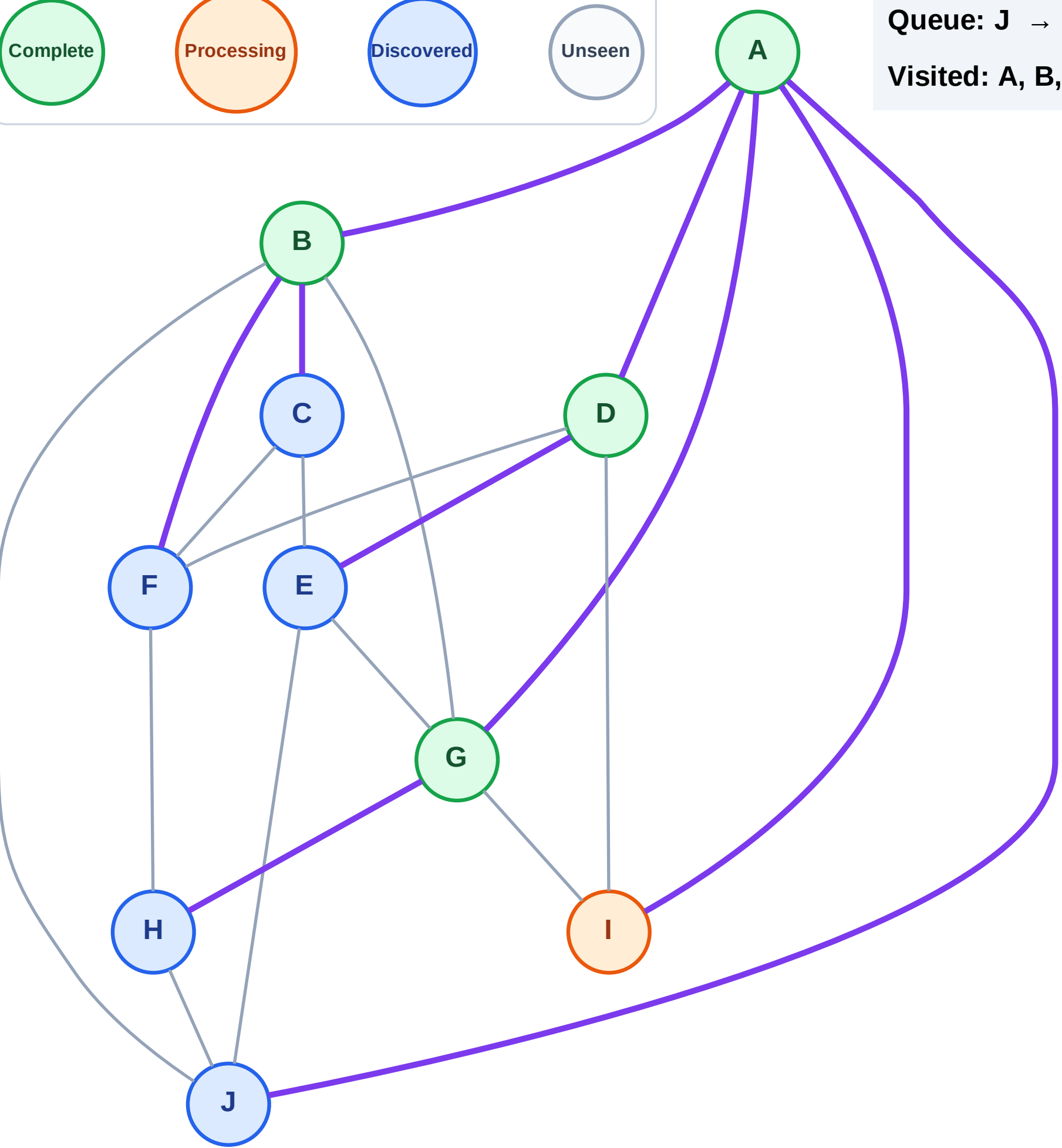
Step 10: Process node I

Legend



Queue: J → C → F → E → H

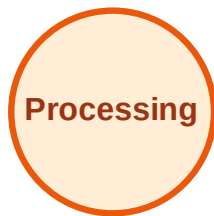
Visited: A, B, D, G



BFS Traversal

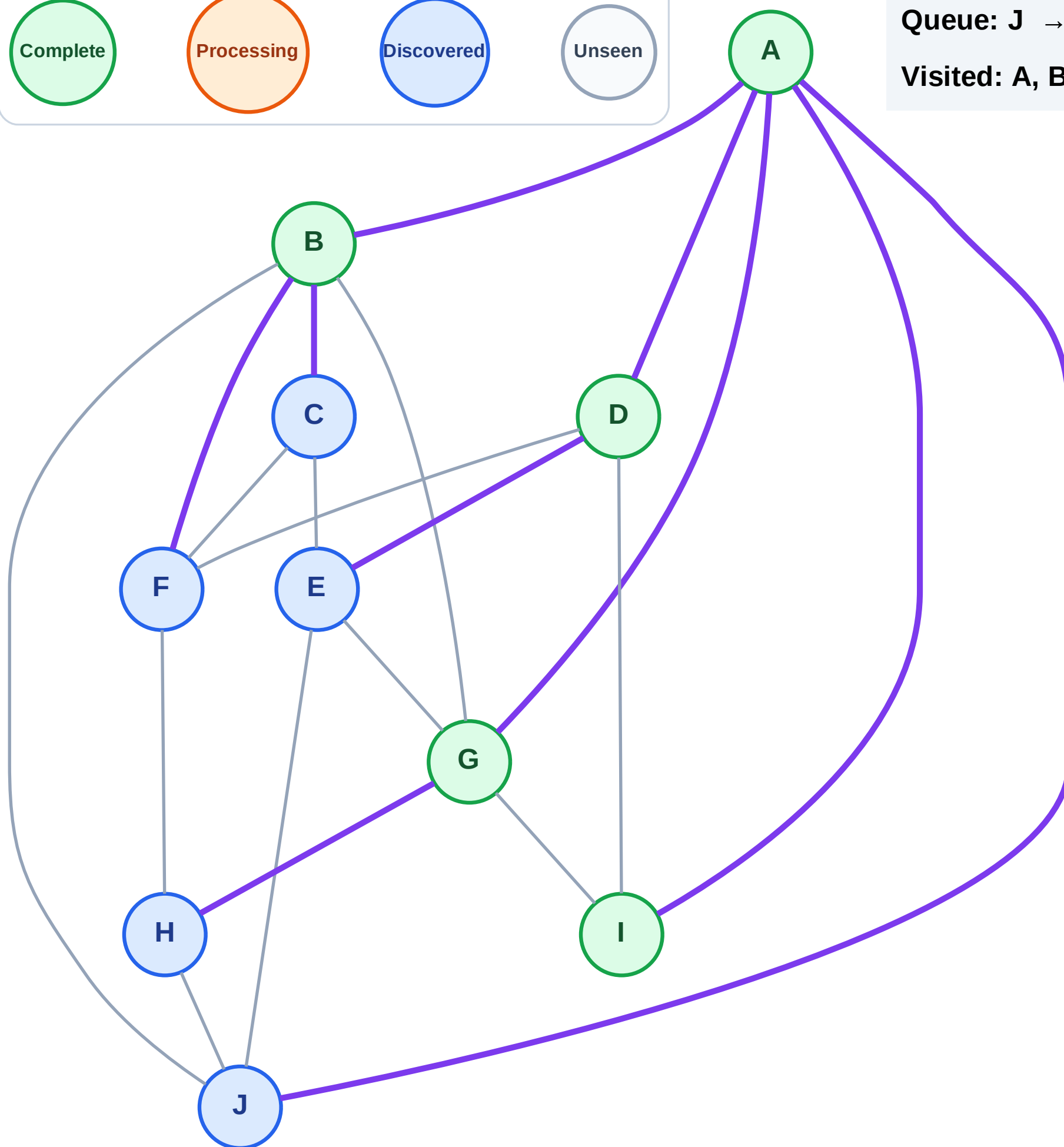
Step 11: No new neighbors from I

Legend



Queue: J → C → F → E → H

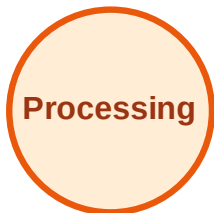
Visited: A, B, D, G, I



BFS Traversal

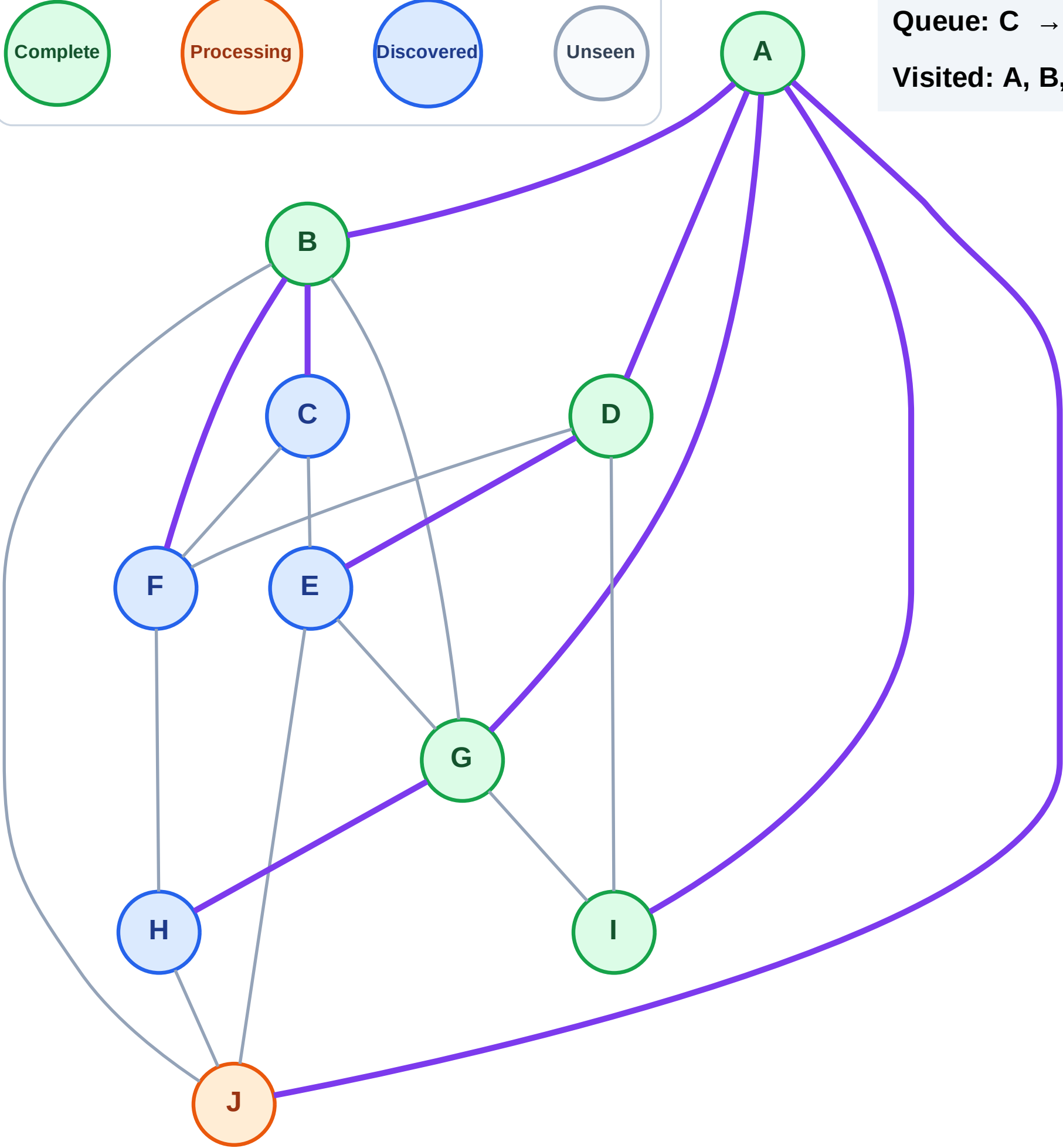
Step 12: Process node J

Legend



Queue: C → F → E → H

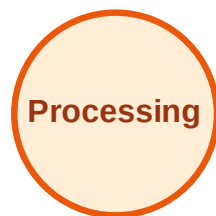
Visited: A, B, D, G, I



BFS Traversal

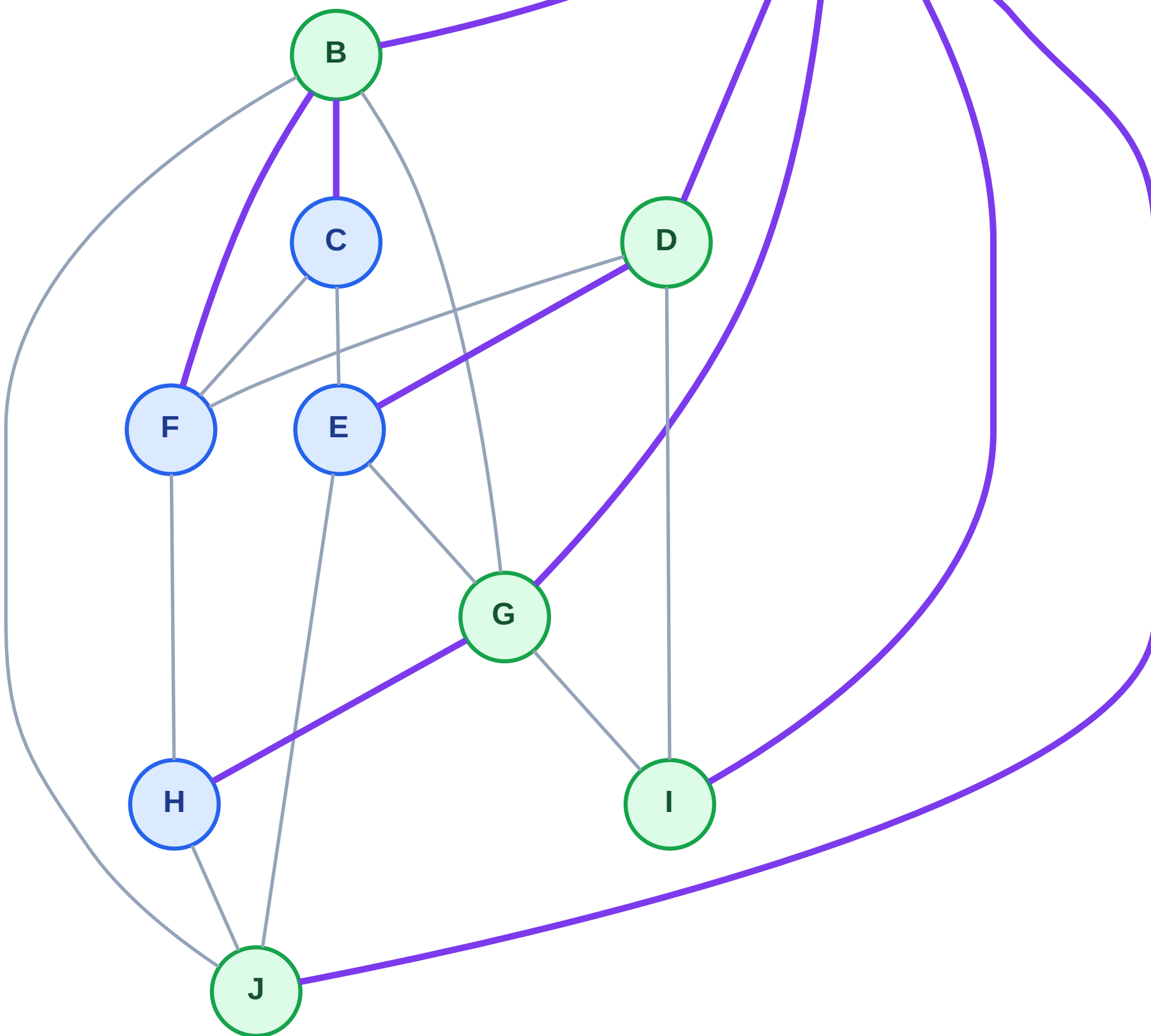
Step 13: No new neighbors from J

Legend



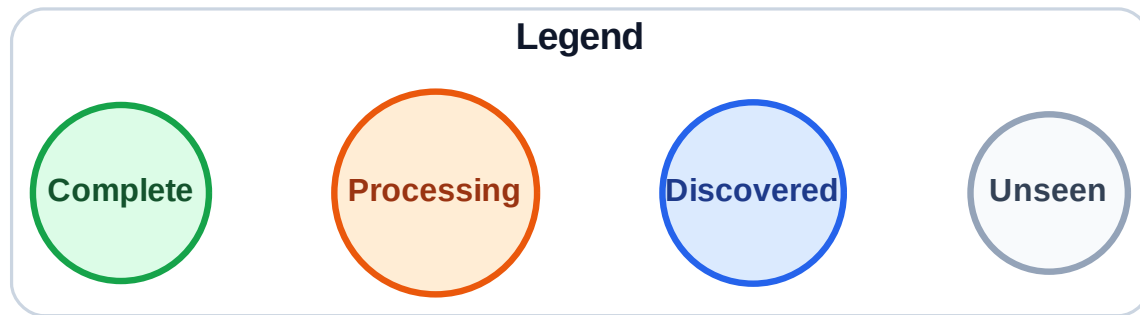
Queue: C → F → E → H

Visited: A, B, D, G, I, J

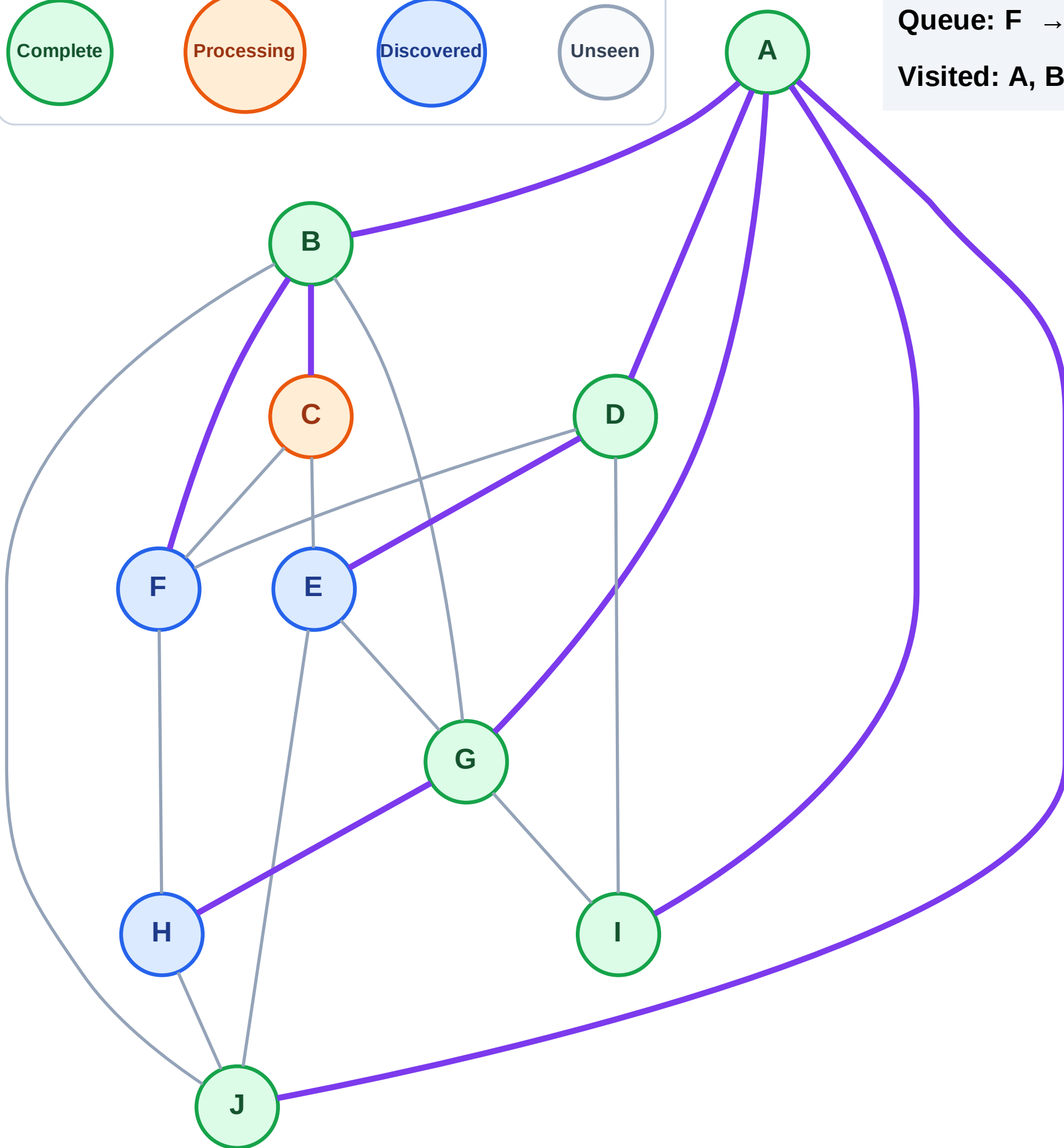


BFS Traversal

Step 14: Process node C



Queue: F → E → H
Visited: A, B, D, G, I, J



BFS Traversal

Step 15: No new neighbors from C

Legend

Complete

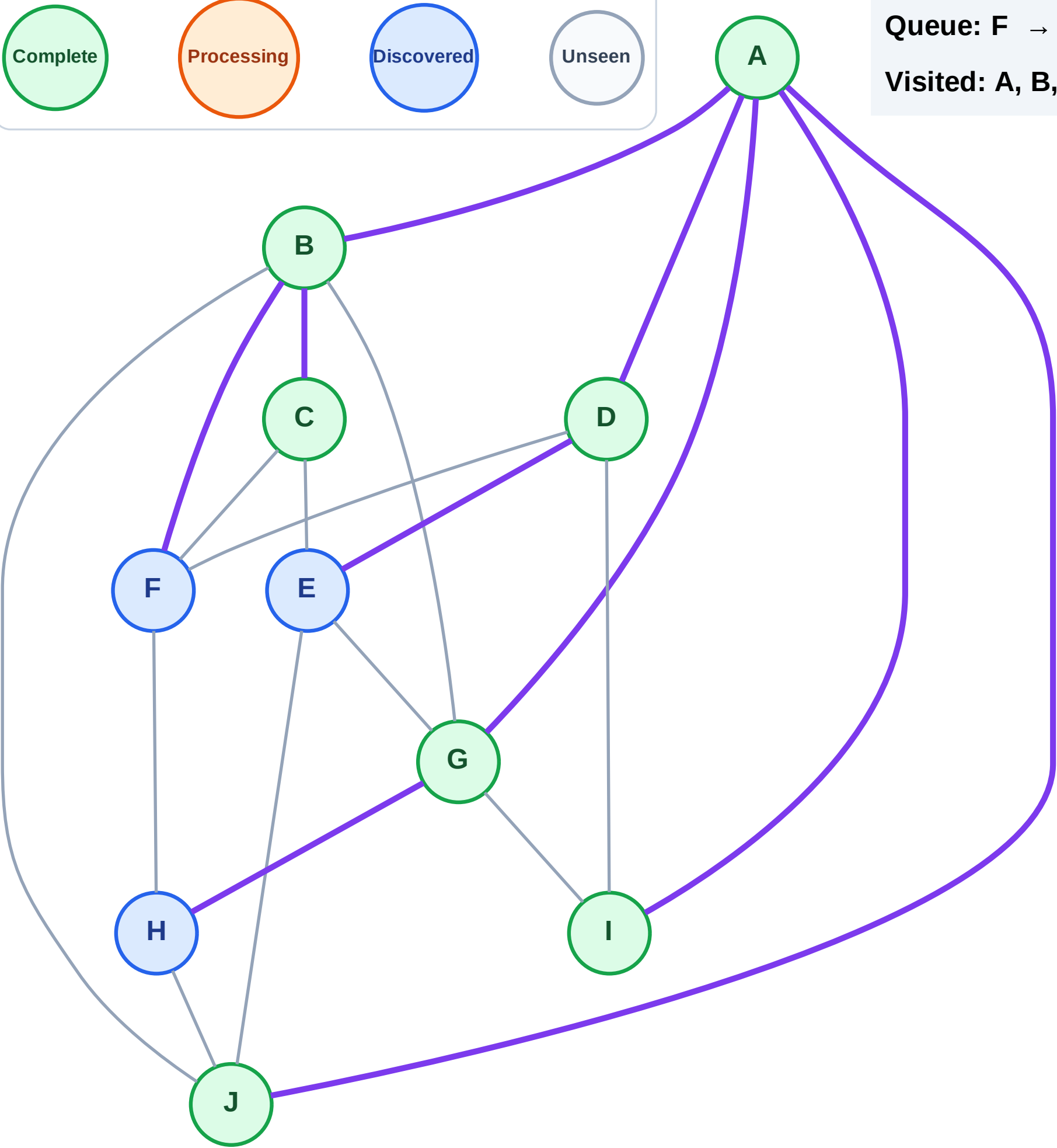
Processing

Discovered

Unseen

Queue: F → E → H

Visited: A, B, C, D, G, I, J



BFS Traversal

Step 16: Process node F

Legend

Complete

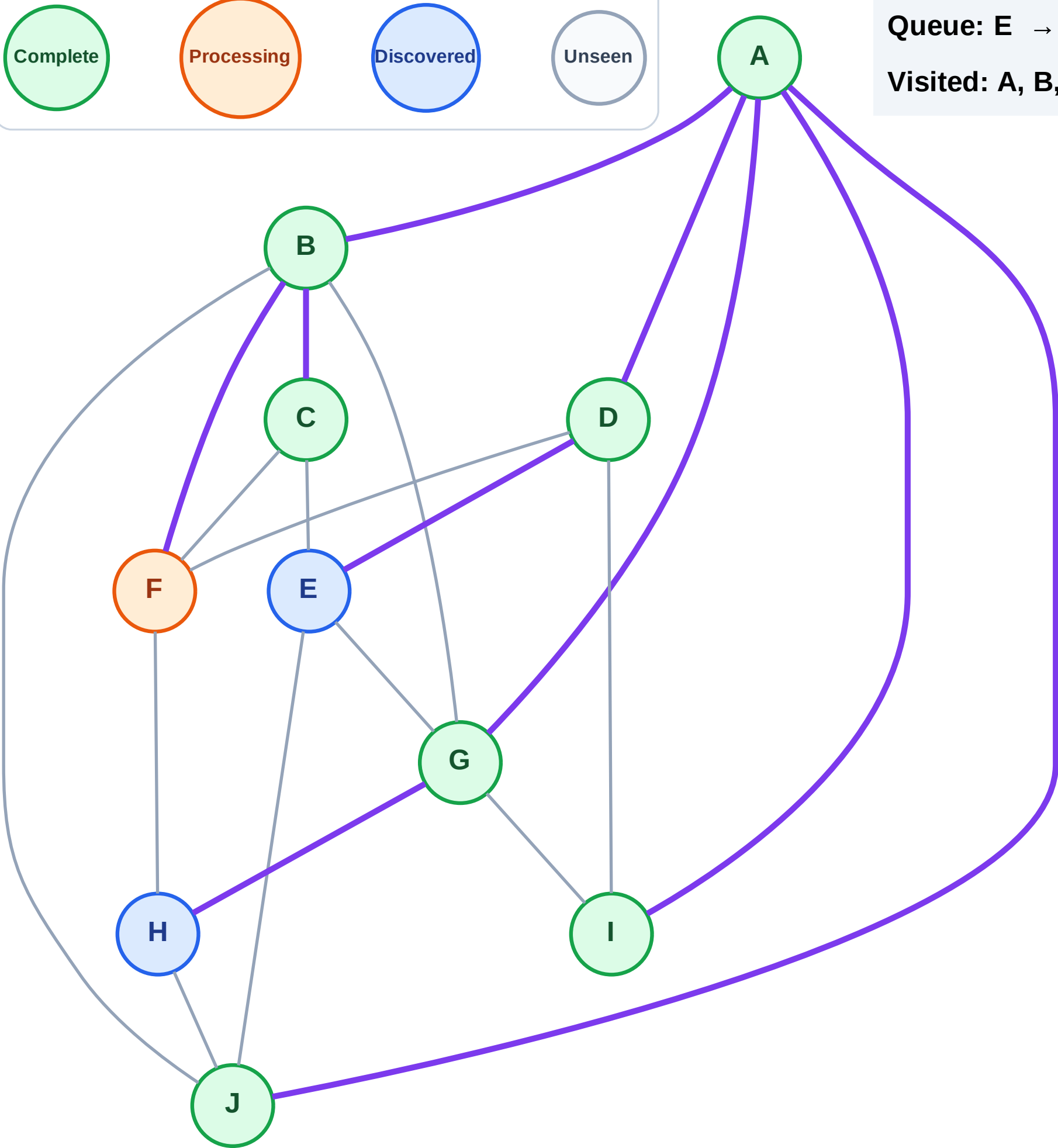
Processing

Discovered

Unseen

Queue: E → H

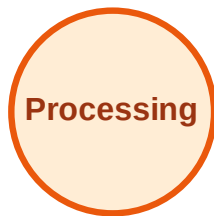
Visited: A, B, C, D, G, I, J



BFS Traversal

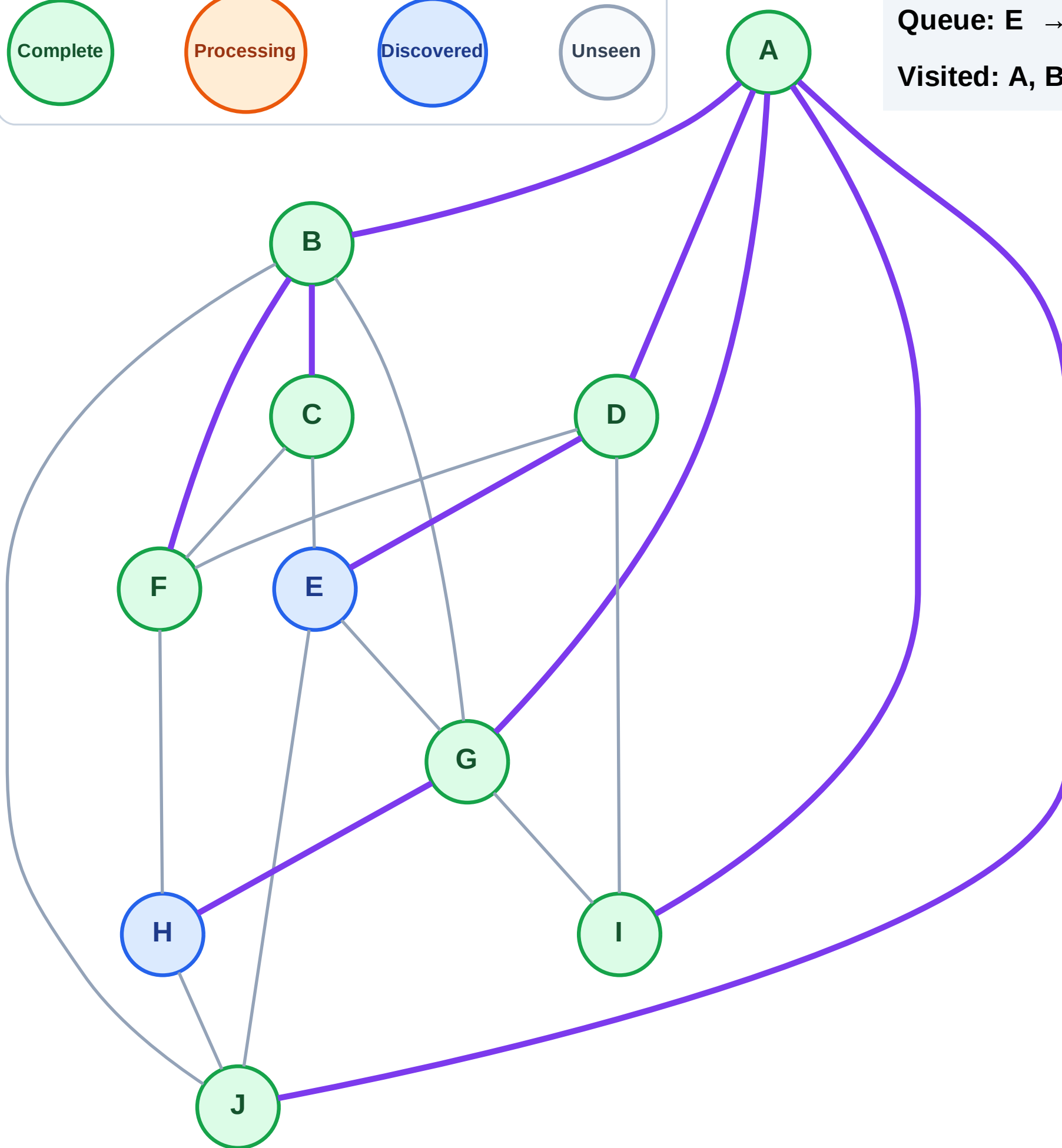
Step 17: No new neighbors from F

Legend



Queue: E → H

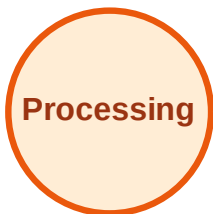
Visited: A, B, C, D, F, G, I, J



BFS Traversal

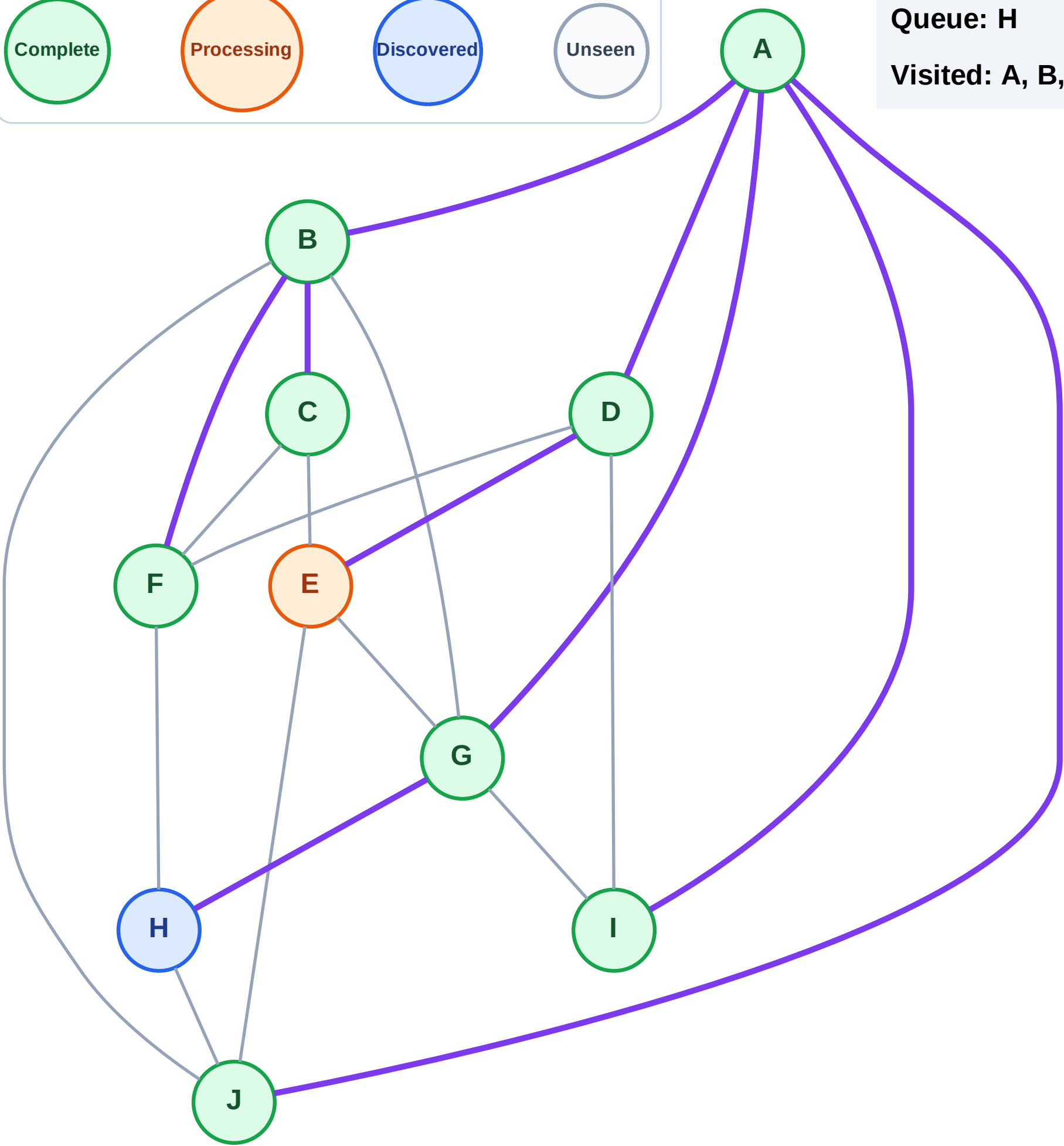
Step 18: Process node E

Legend



Queue: H

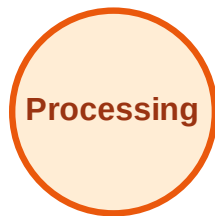
Visited: A, B, C, D, F, G, I, J



BFS Traversal

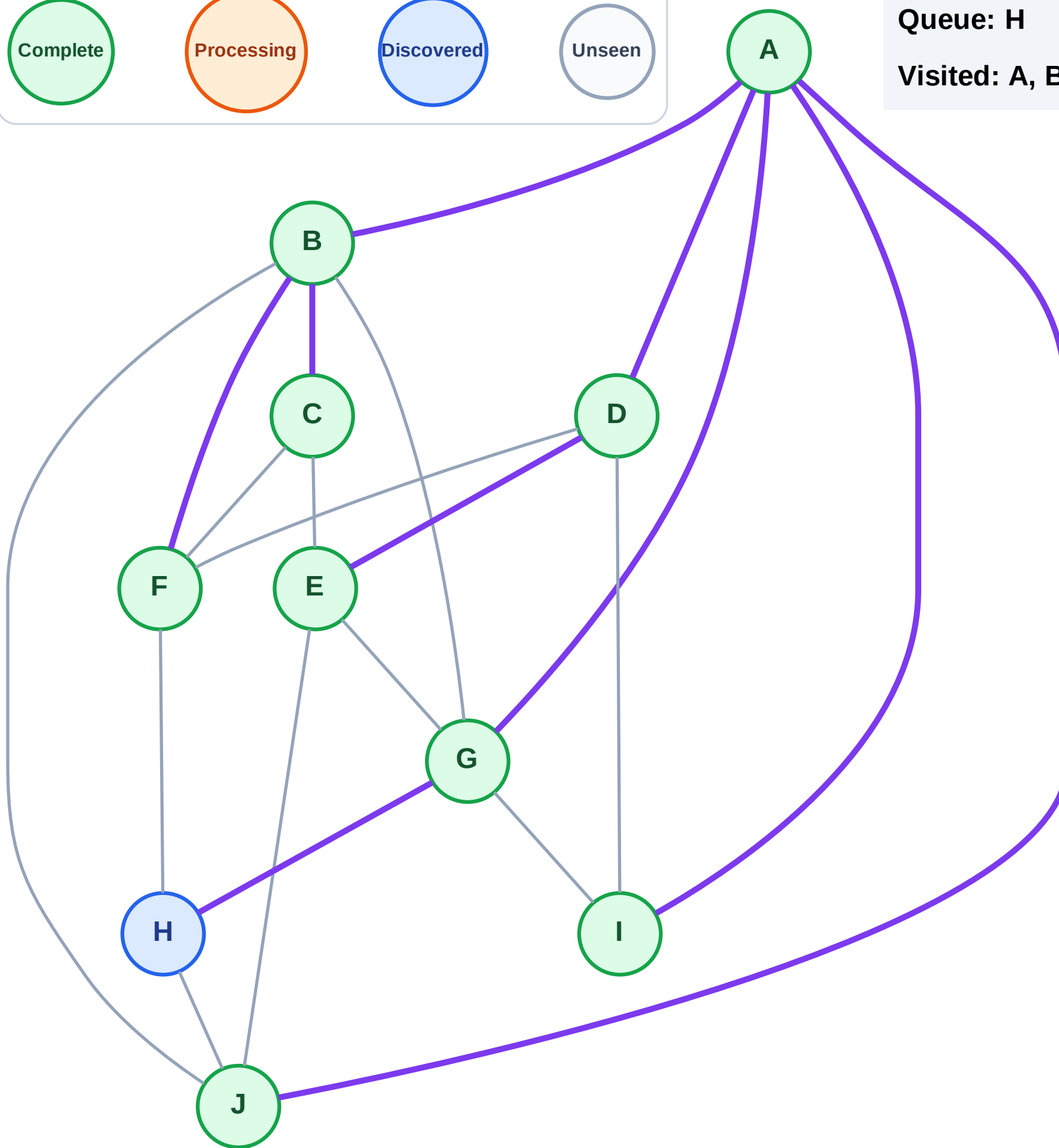
Step 19: No new neighbors from E

Legend



Queue: H

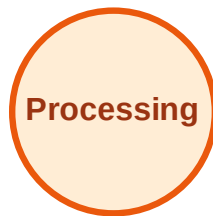
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

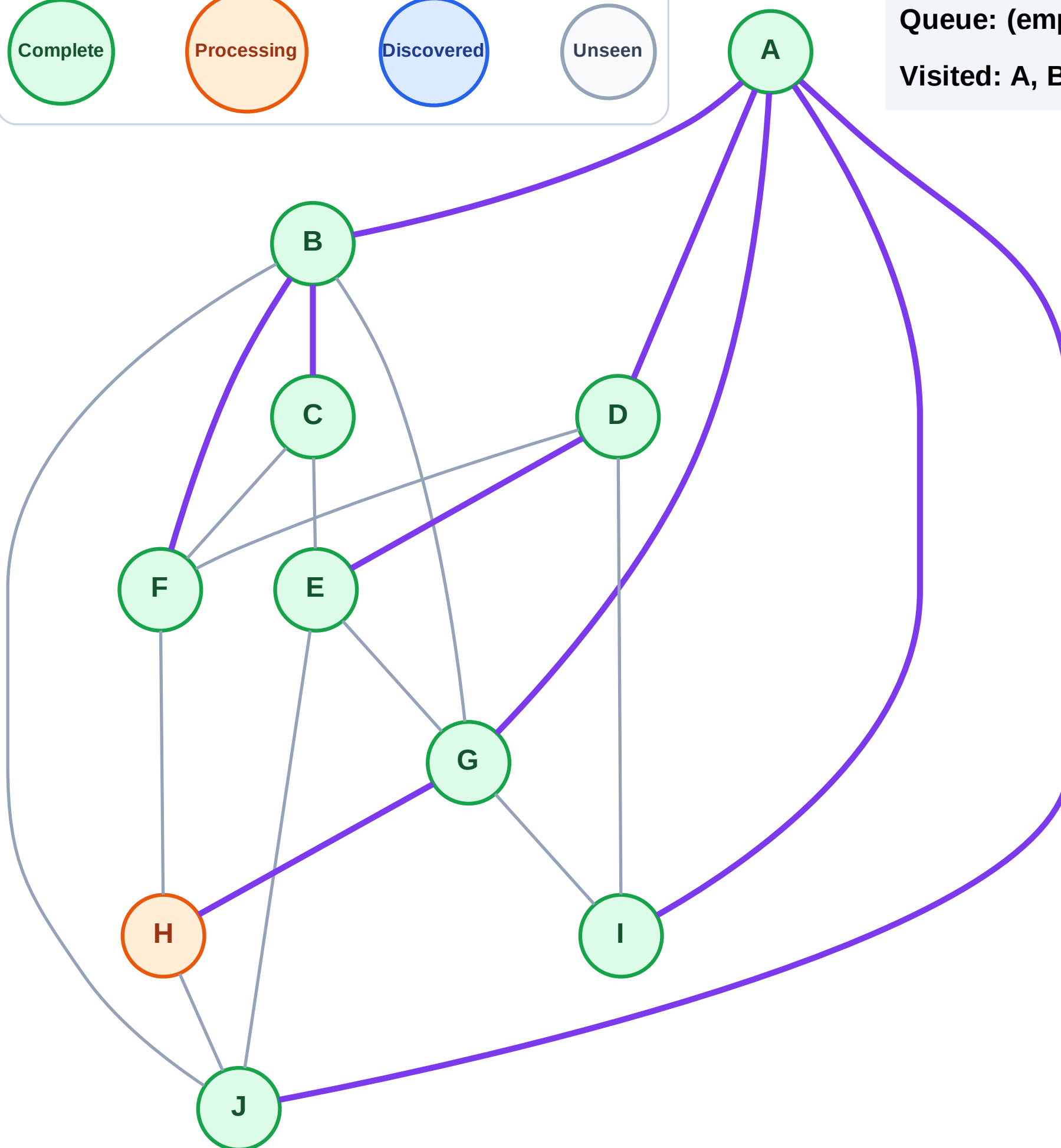
Step 20: Process node H

Legend



Queue: (empty)

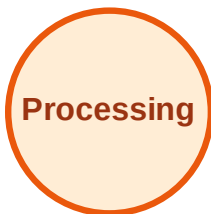
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

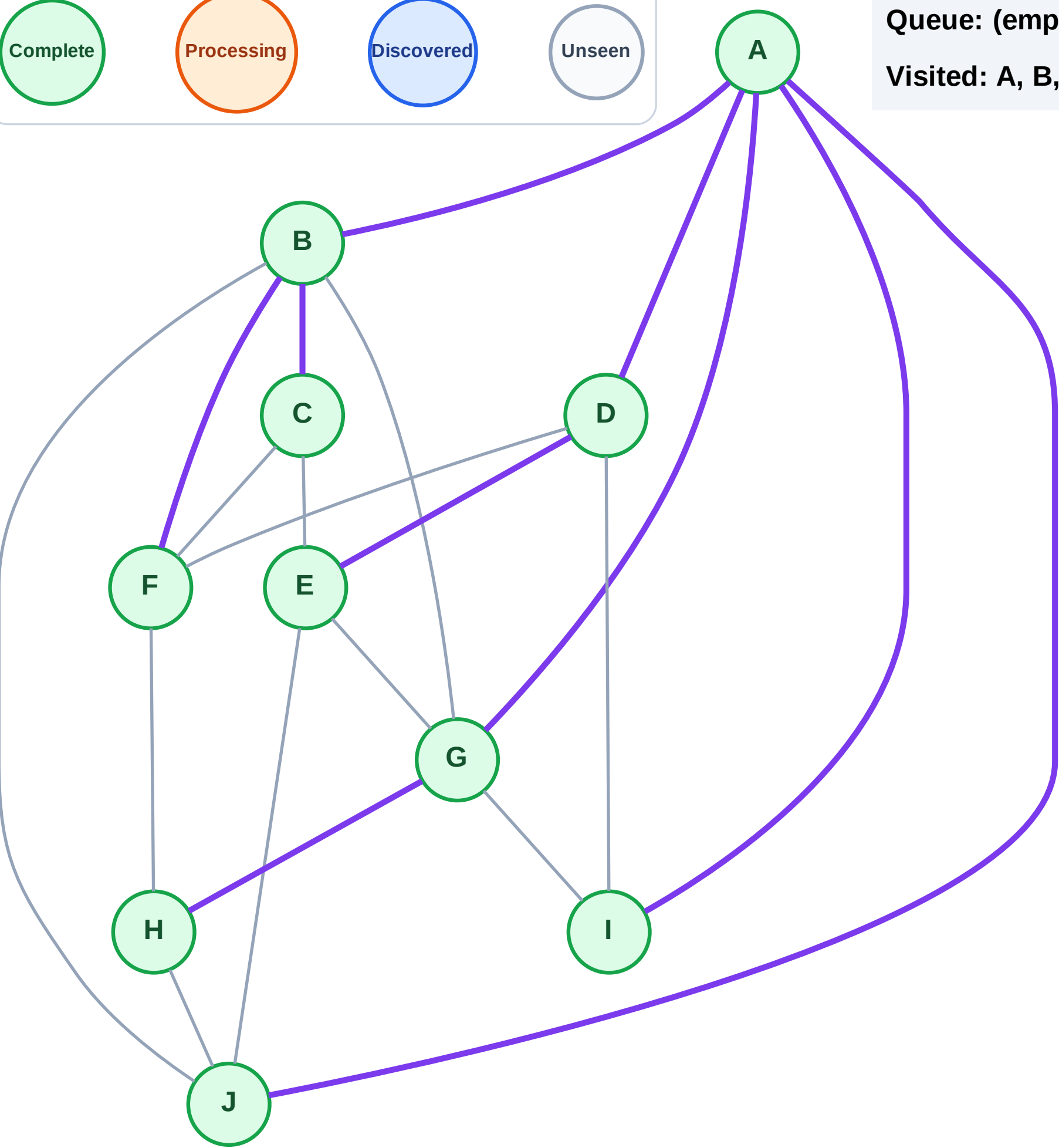
Step 21: No new neighbors from H

Legend



Queue: (empty)

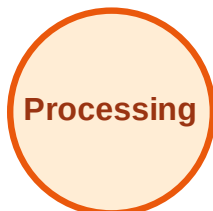
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

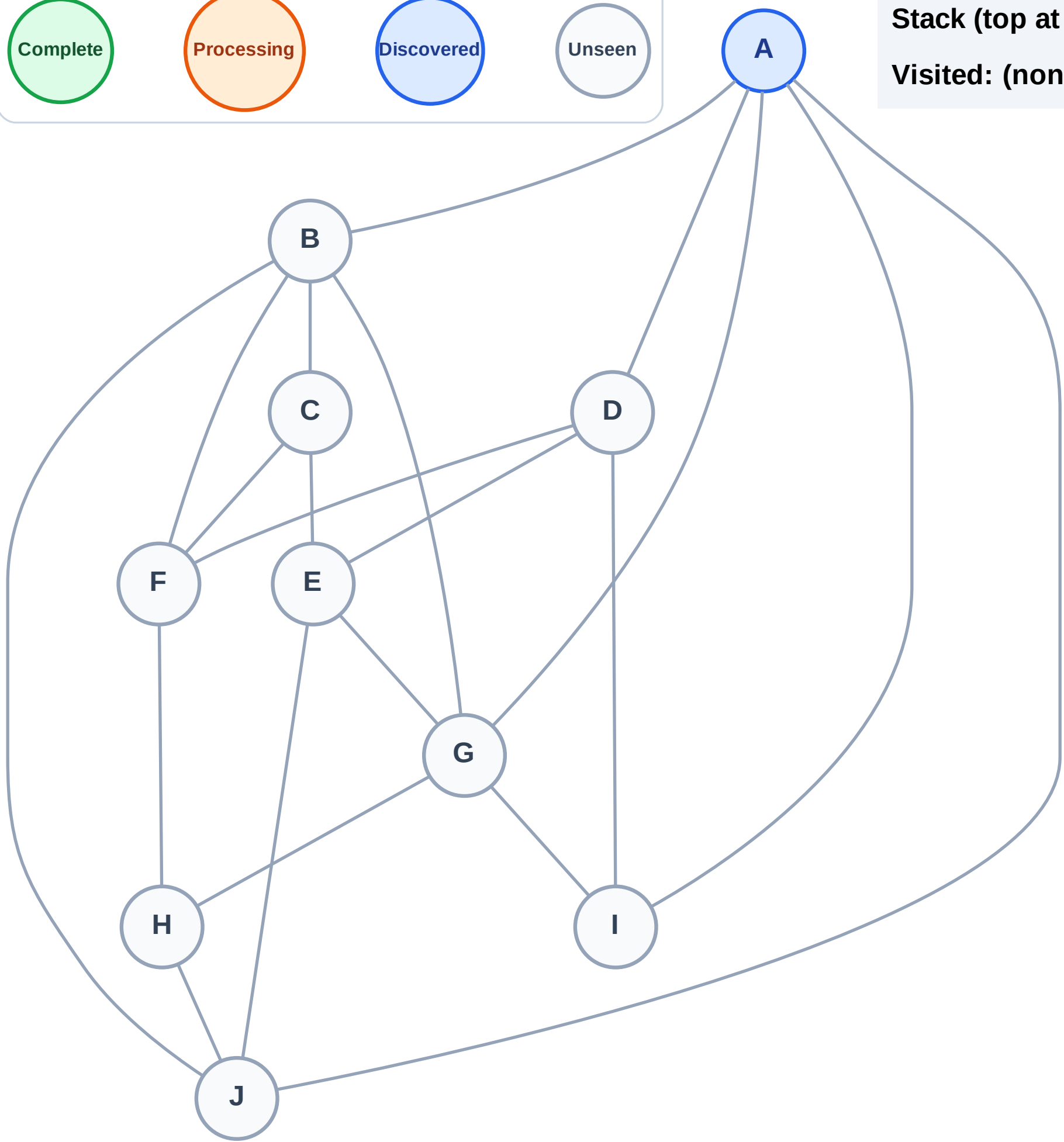
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

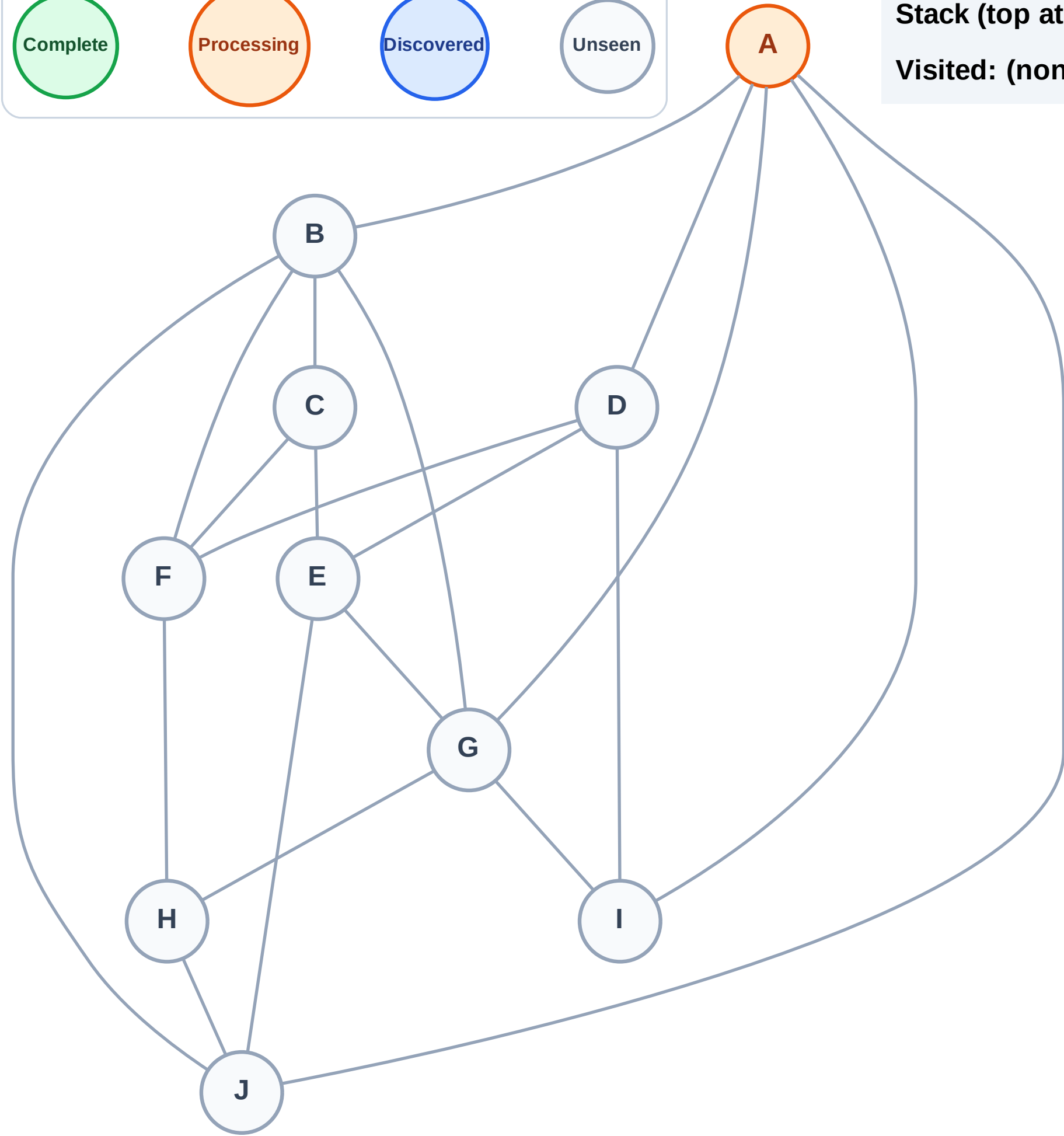
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



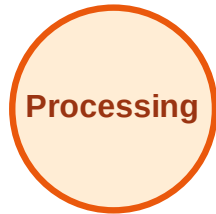
DFS Traversal

Step 3: Discover J, I, G, D, B from A

Legend



Complete



Processing



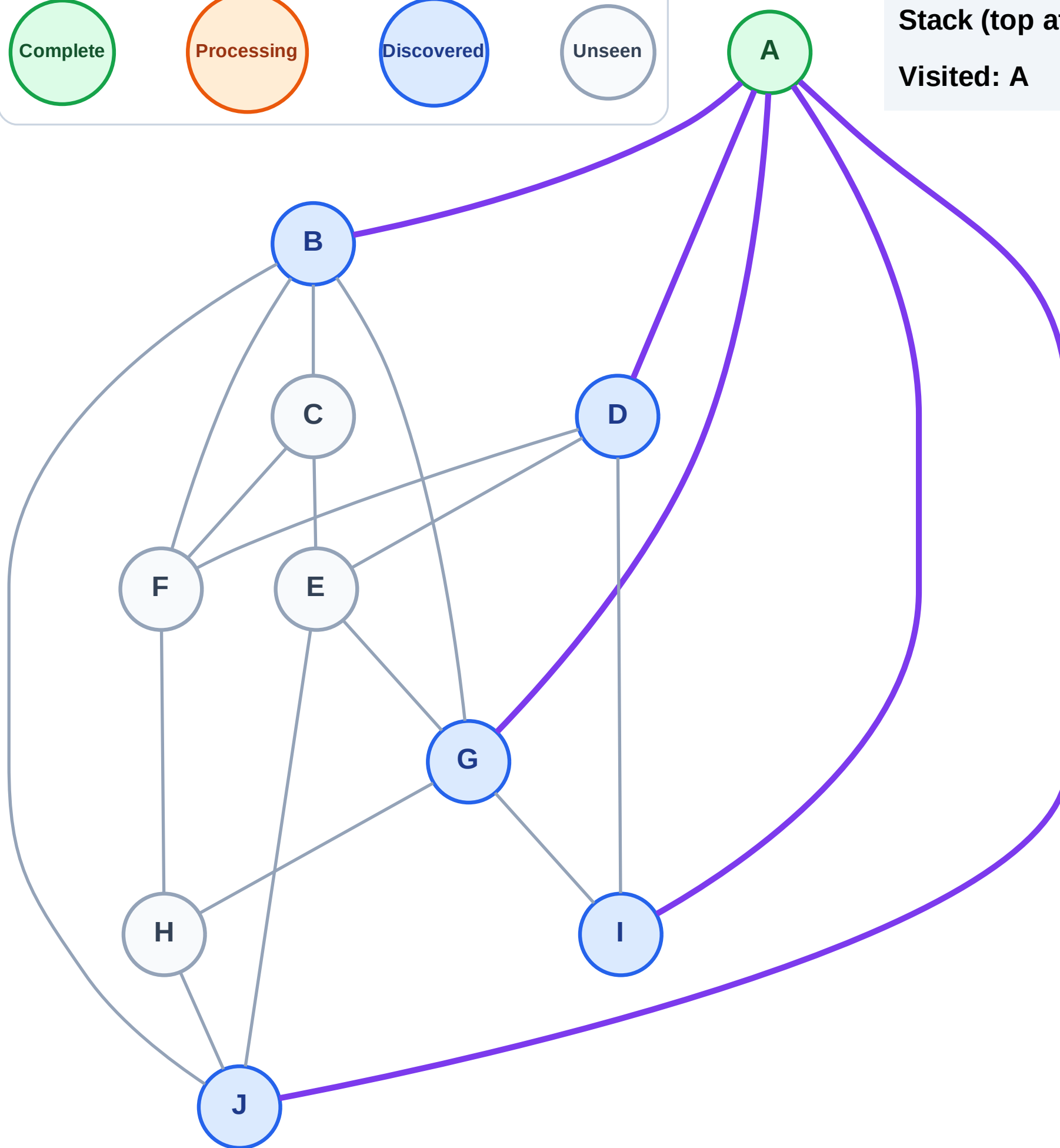
Discovered



Unseen

Stack (top at right): J | I | G | D | B

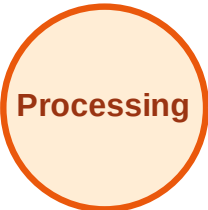
Visited: A



DFS Traversal

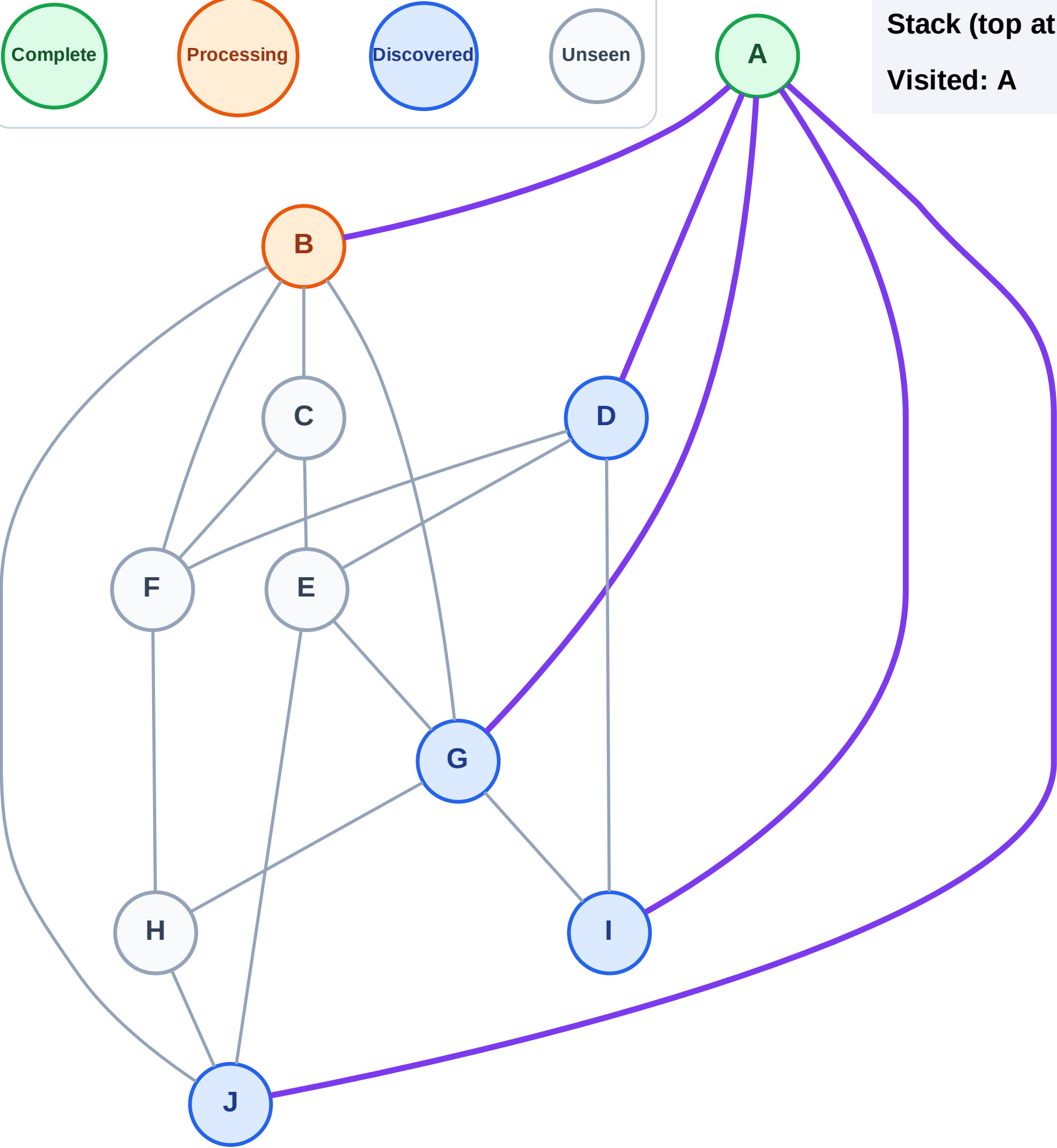
Step 4: Process node B

Legend



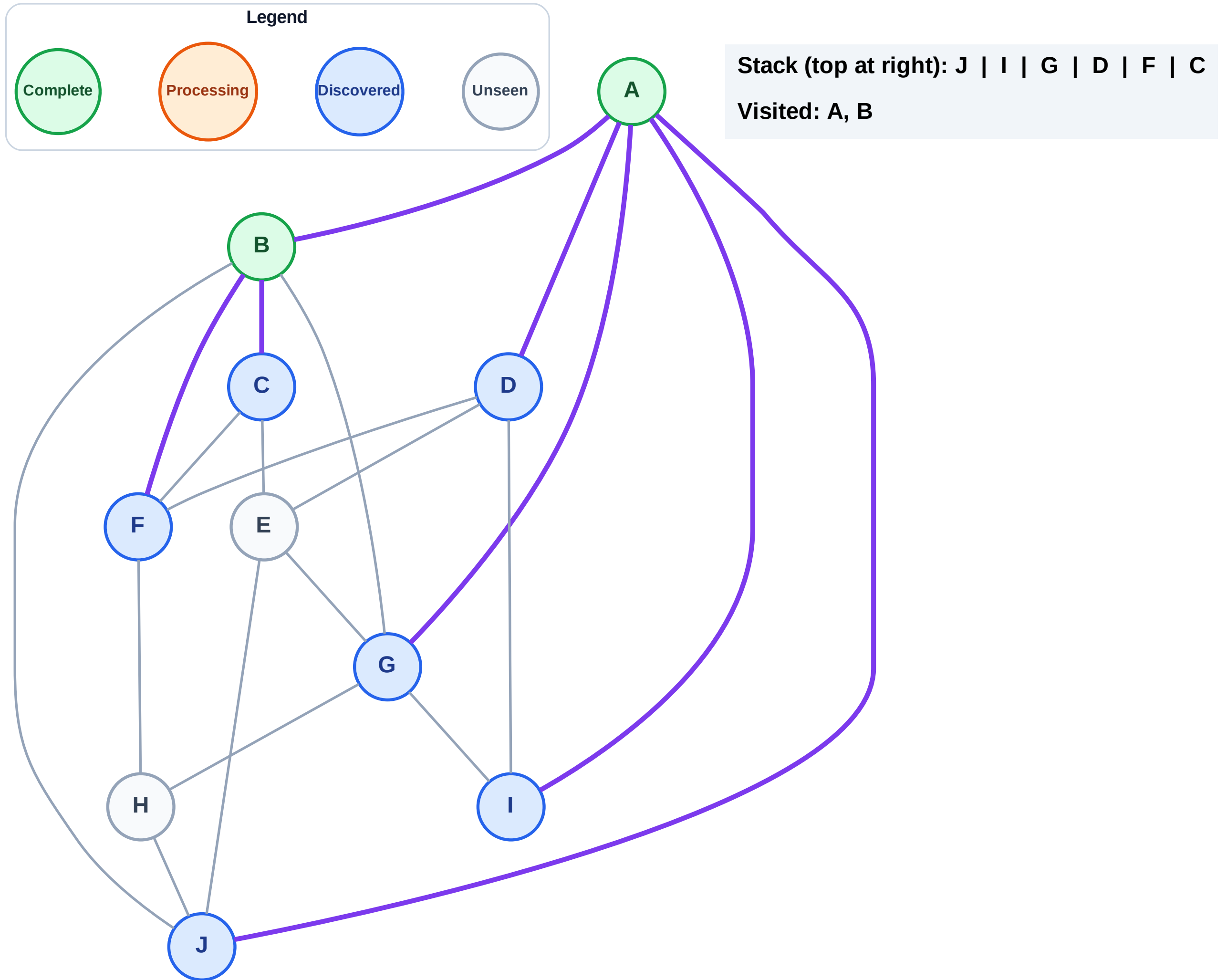
Stack (top at right): J | I | G | D

Visited: A

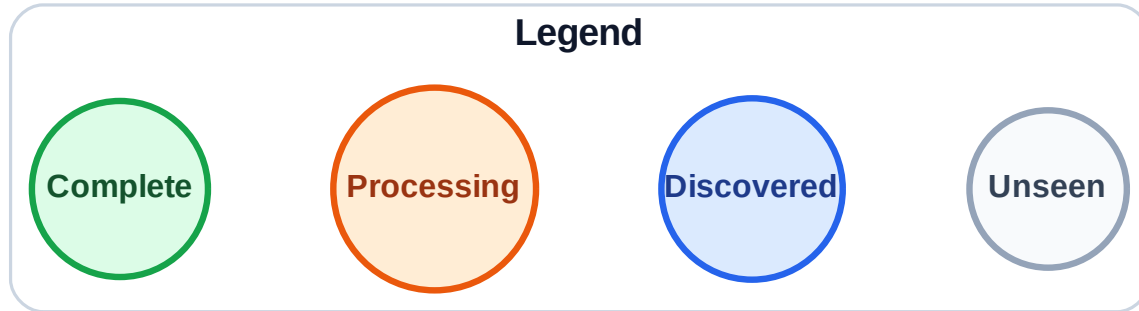


Step 5: Discover F, C from B

Step 5: Discover F, C from B

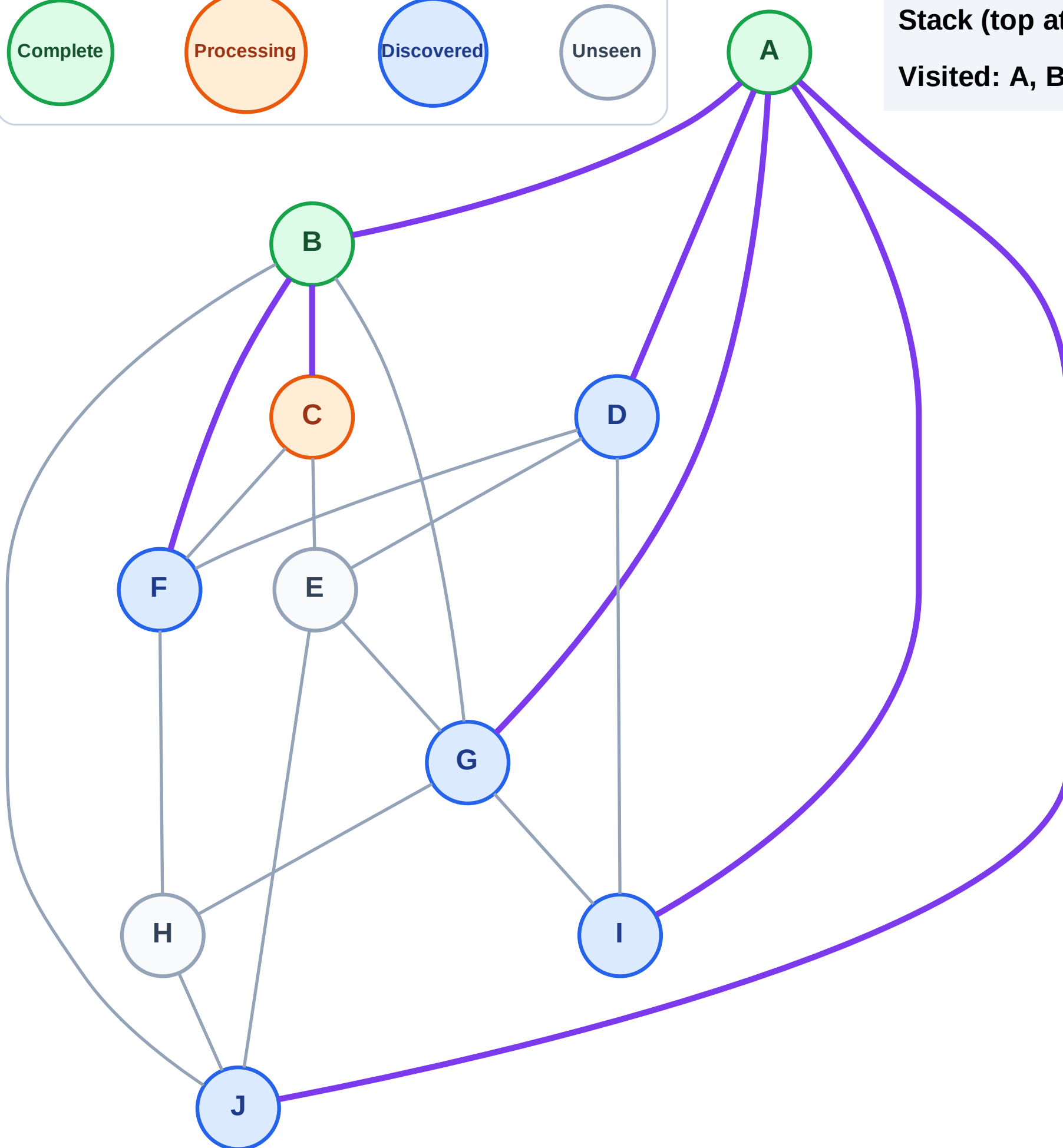


Step 6: Process node C



Stack (top at right): J | I | G | D | F

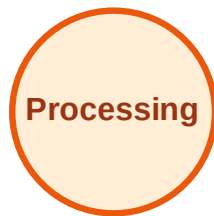
Visited: A, B



DFS Traversal

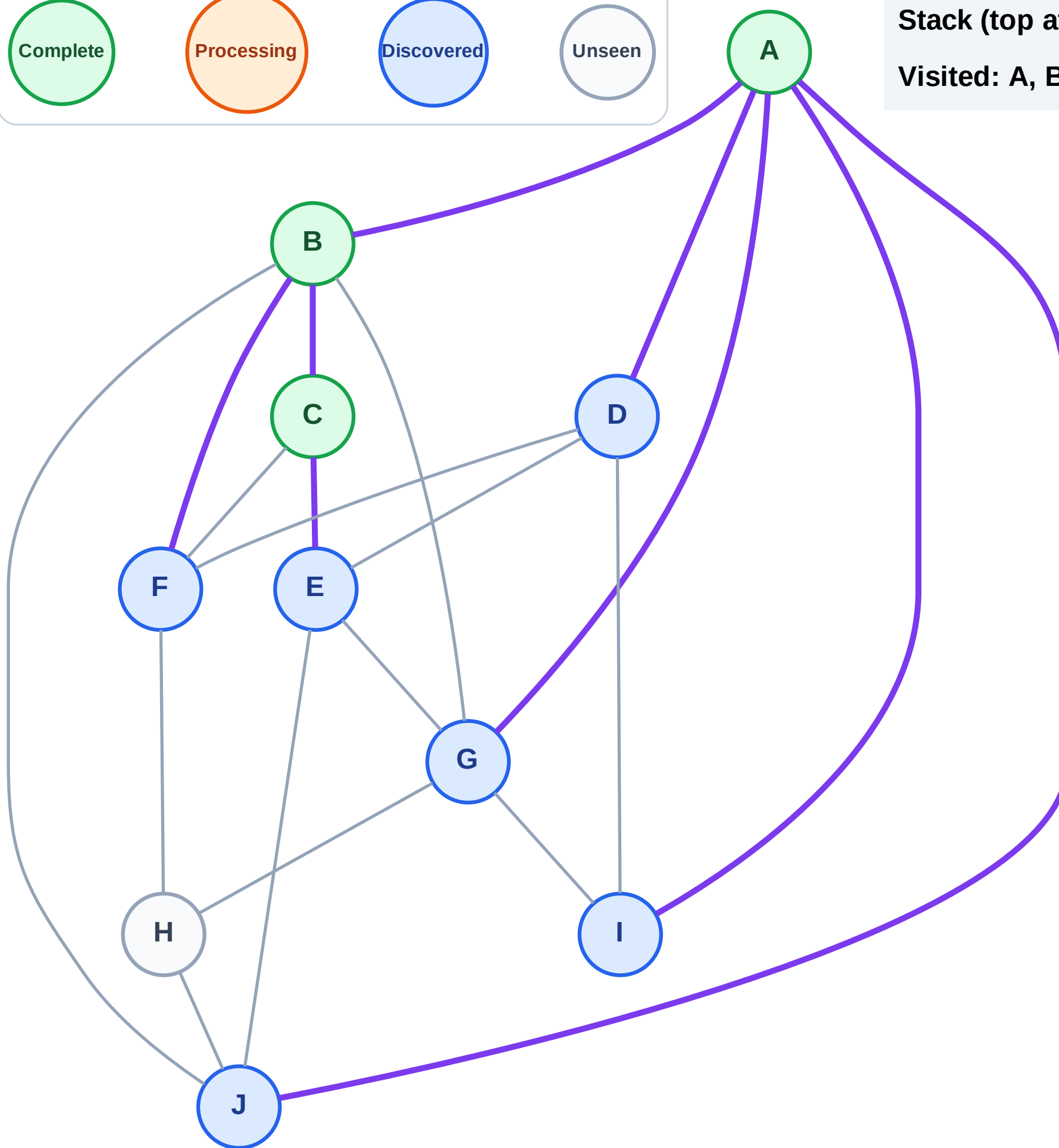
Step 7: Discover E from C

Legend



Stack (top at right): J | I | G | D | F | E

Visited: A, B, C



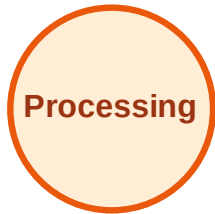
DFS Traversal

Step 8: Process node E

Legend



Complete



Processing



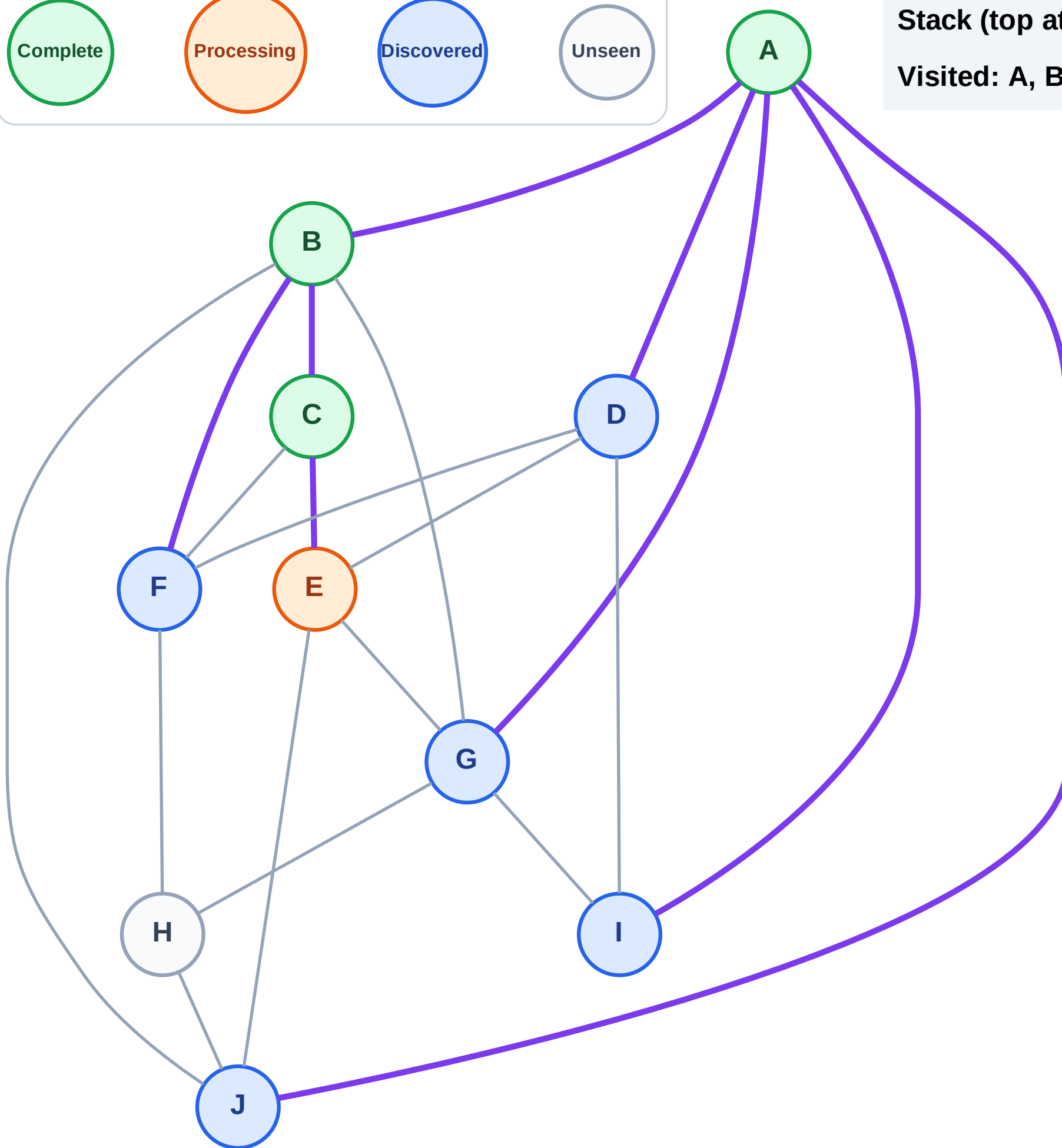
Discovered



Unseen

Stack (top at right): J | I | G | D | F

Visited: A, B, C



DFS Traversal

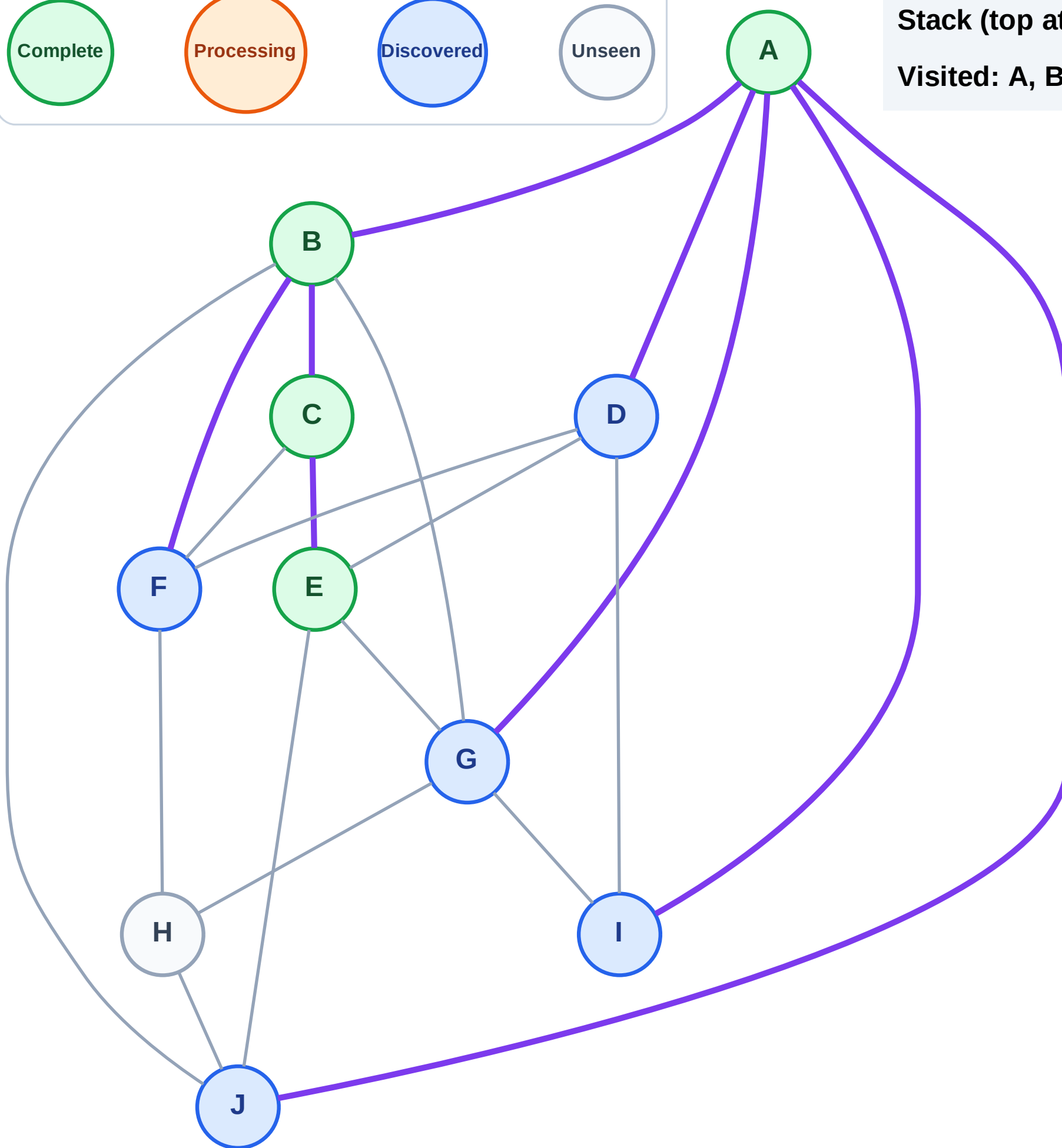
Step 9: No new neighbors from E

Legend



Stack (top at right): J | I | G | D | F

Visited: A, B, C, E



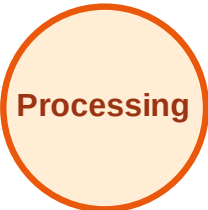
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



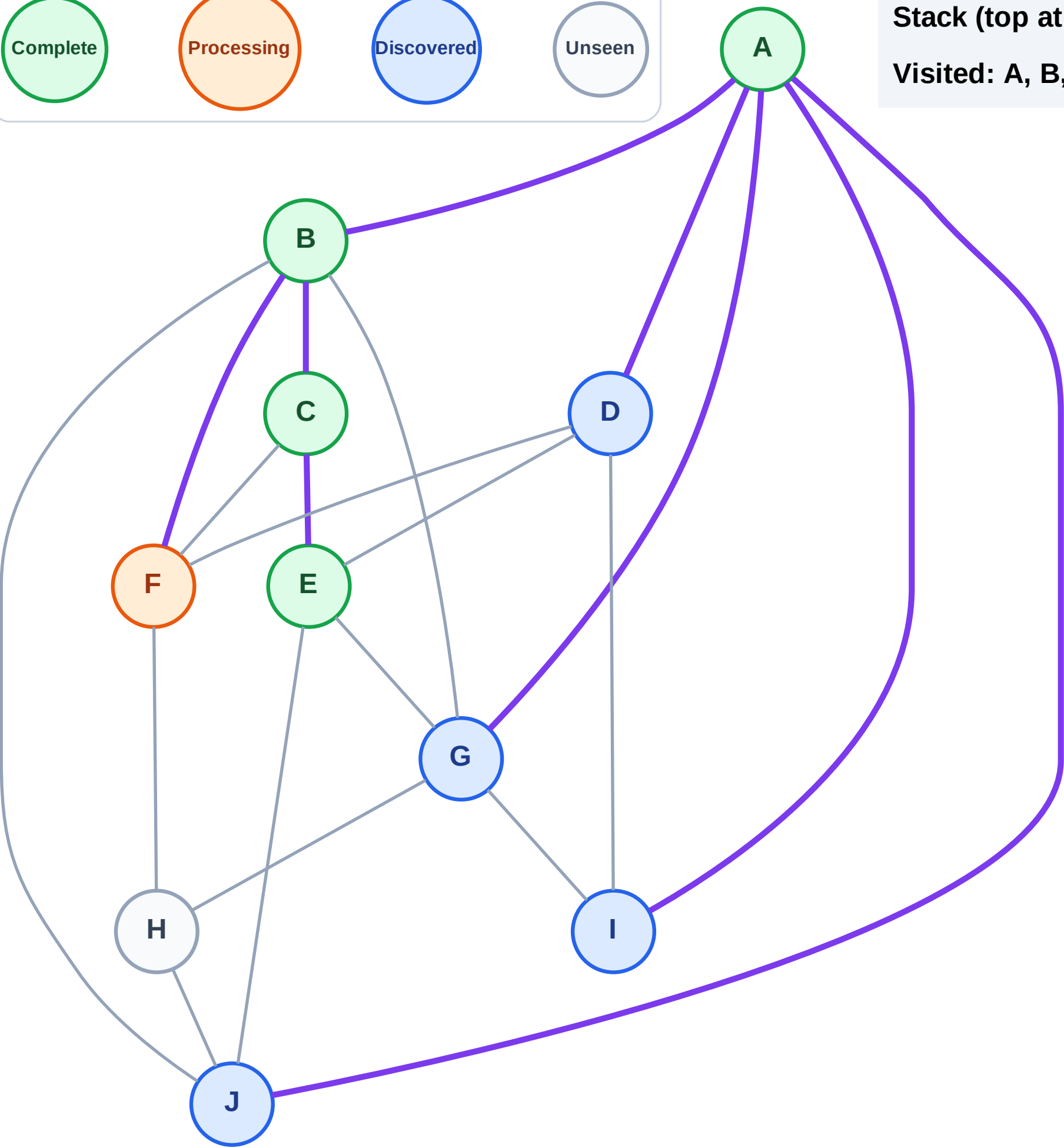
Discovered



Unseen

Stack (top at right): J | I | G | D

Visited: A, B, C, E



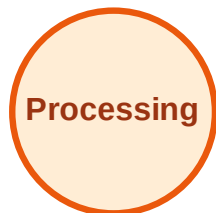
DFS Traversal

Step 11: Discover H from F

Legend



Complete



Processing



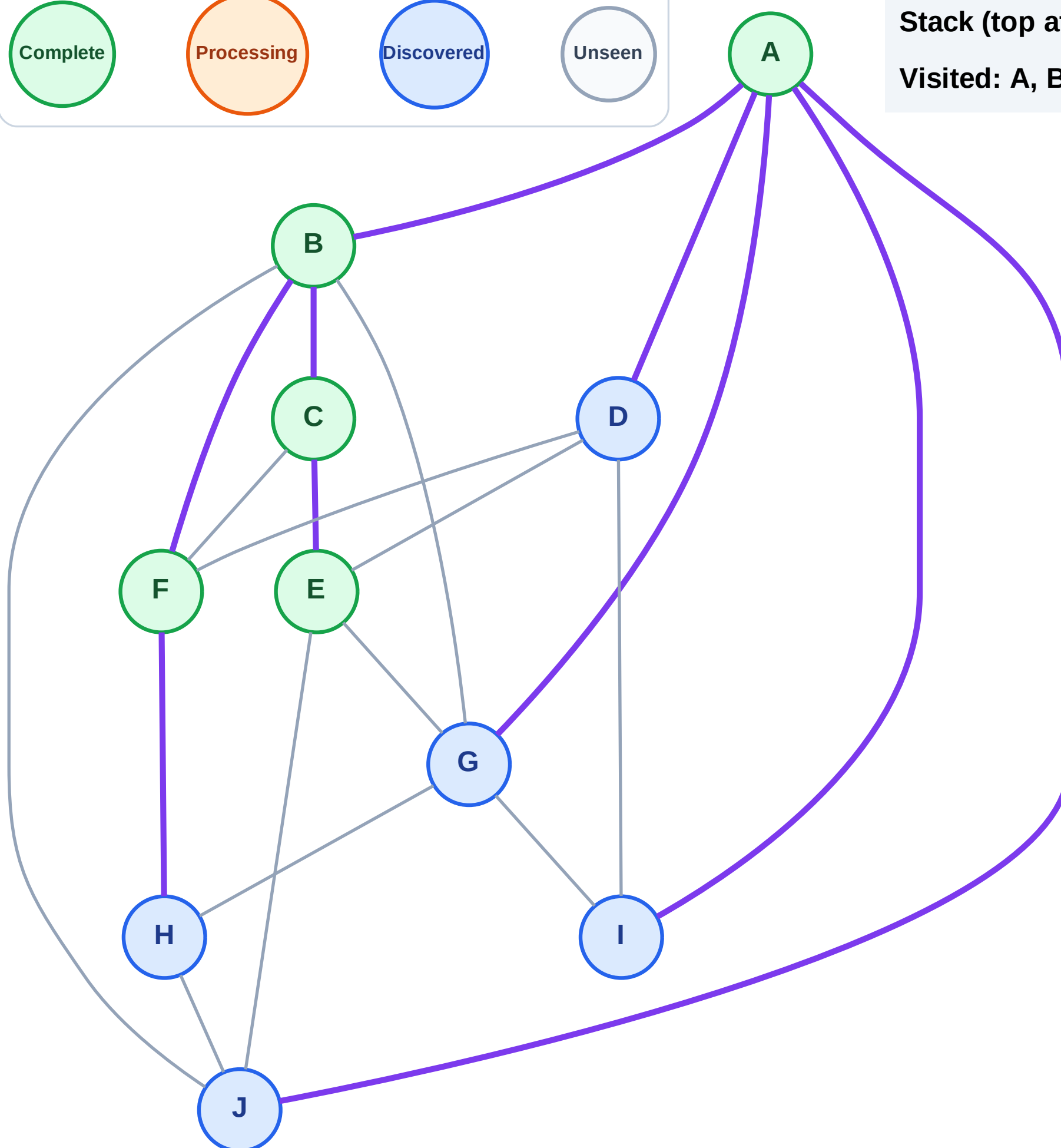
Discovered



Unseen

Stack (top at right): J | I | G | D | H

Visited: A, B, C, E, F



DFS Traversal

Step 12: Process node H

Legend



Complete



Processing



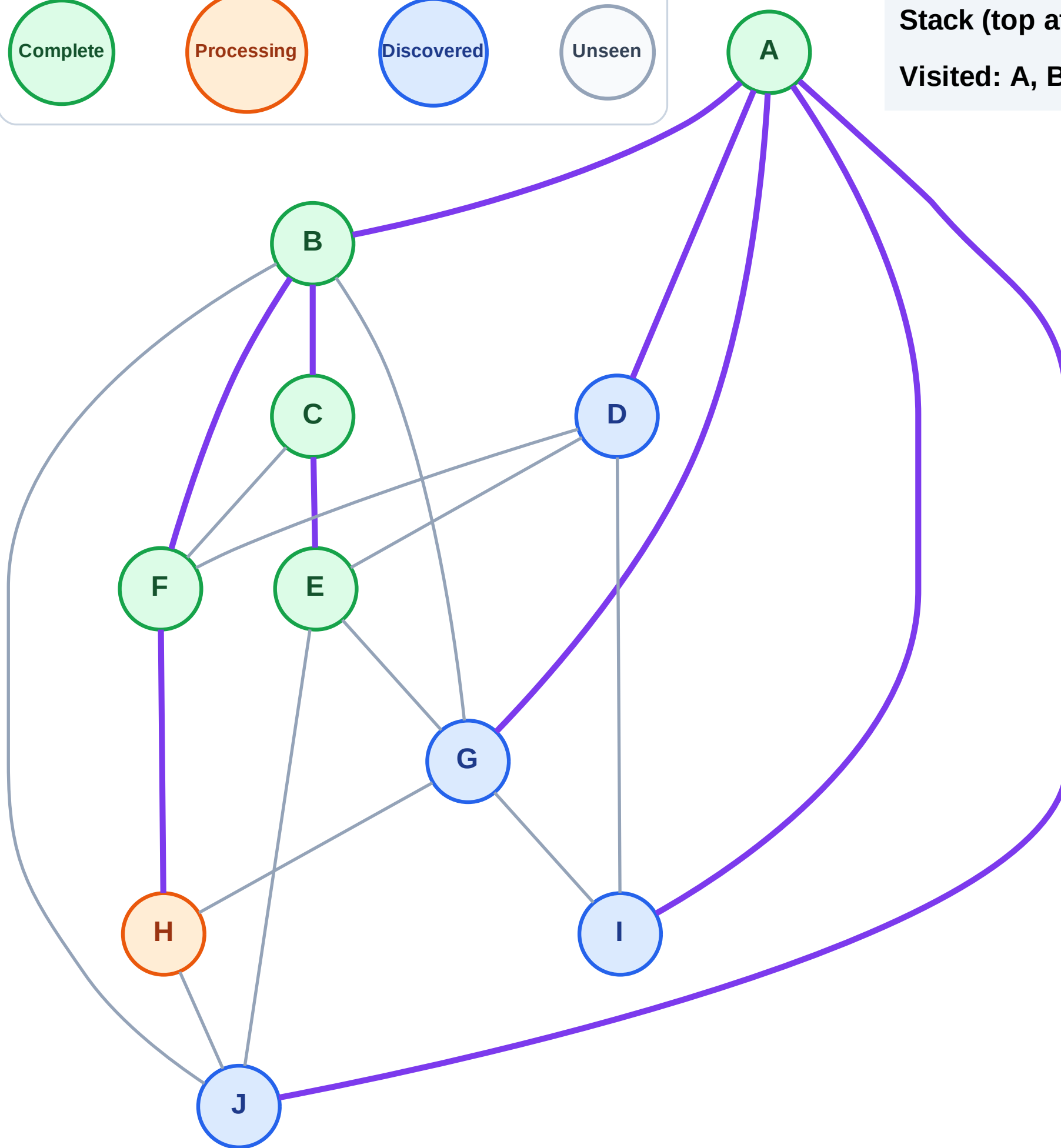
Discovered



Unseen

Stack (top at right): J | I | G | D

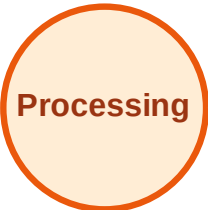
Visited: A, B, C, E, F



DFS Traversal

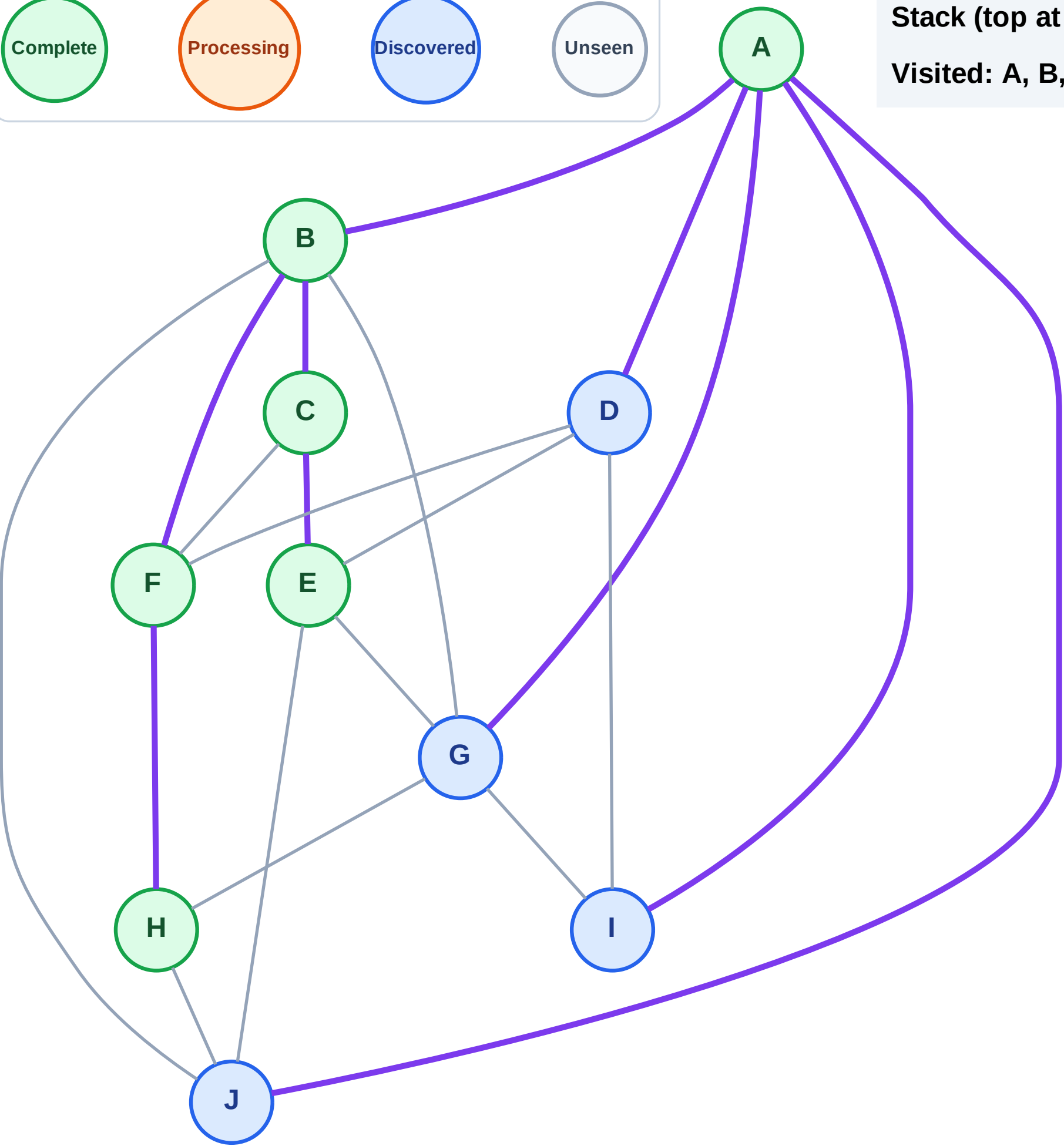
Step 13: No new neighbors from H

Legend



Stack (top at right): J | I | G | D

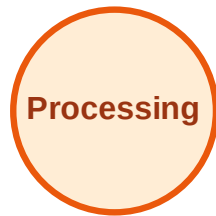
Visited: A, B, C, E, F, H



DFS Traversal

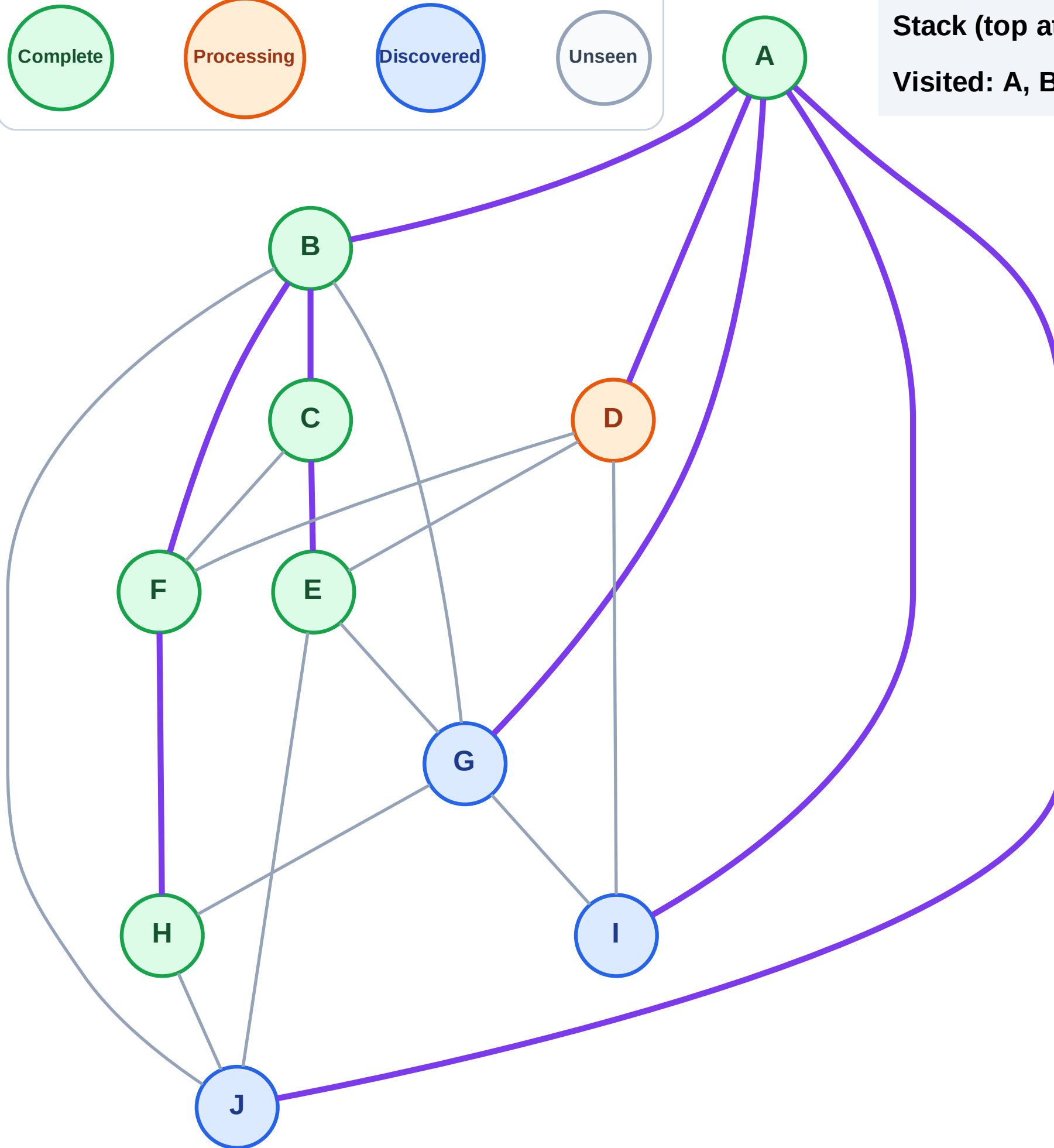
Step 14: Process node D

Legend



Stack (top at right): J | I | G

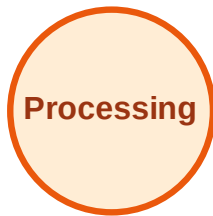
Visited: A, B, C, E, F, H



DFS Traversal

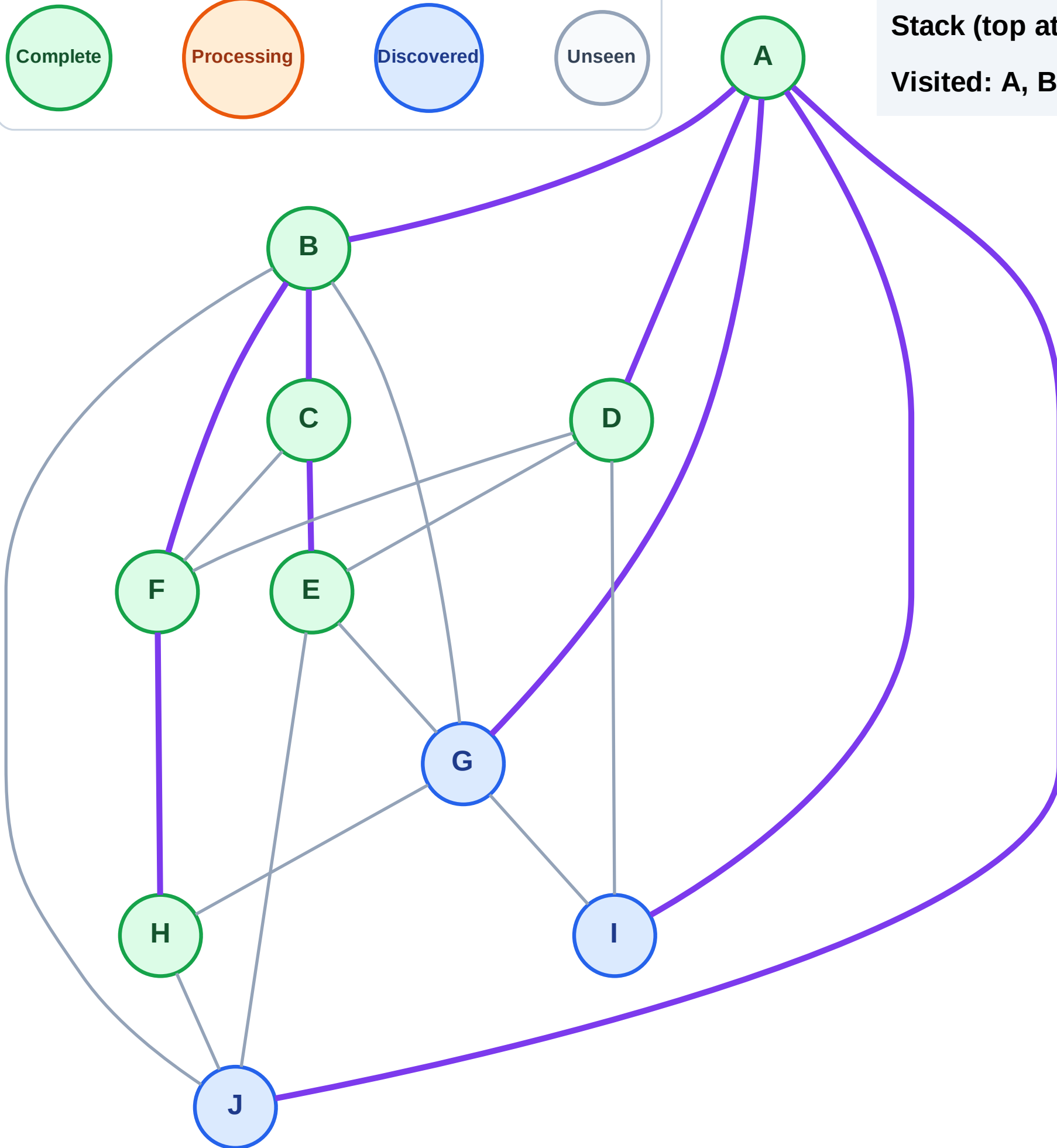
Step 15: No new neighbors from D

Legend



Stack (top at right): J | I | G

Visited: A, B, C, D, E, F, H



DFS Traversal

Step 16: Process node G

Legend

Complete

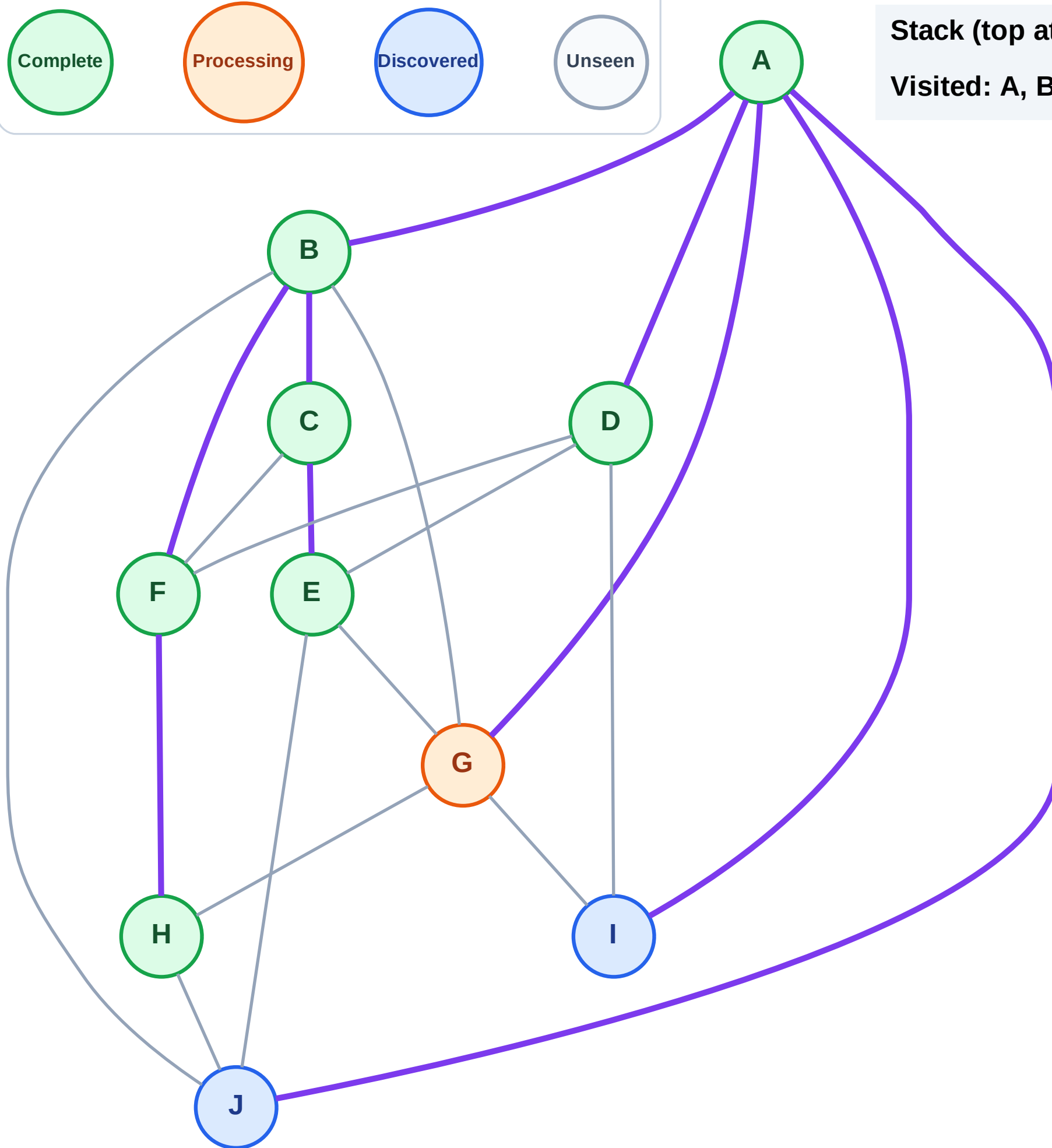
Processing

Discovered

Unseen

Stack (top at right): J | I

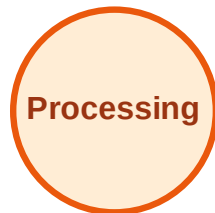
Visited: A, B, C, D, E, F, H



DFS Traversal

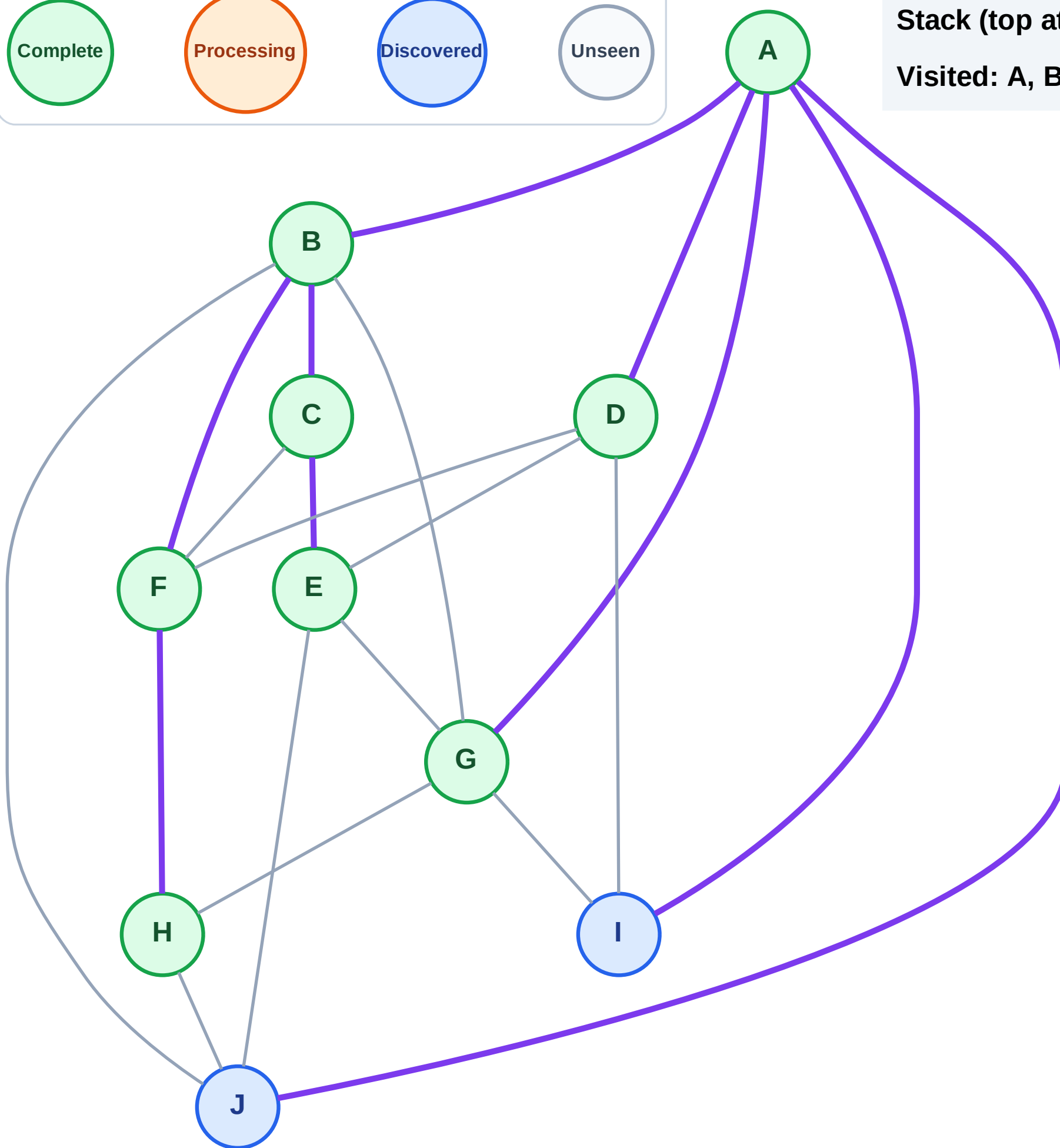
Step 17: No new neighbors from G

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node I

Legend

Complete

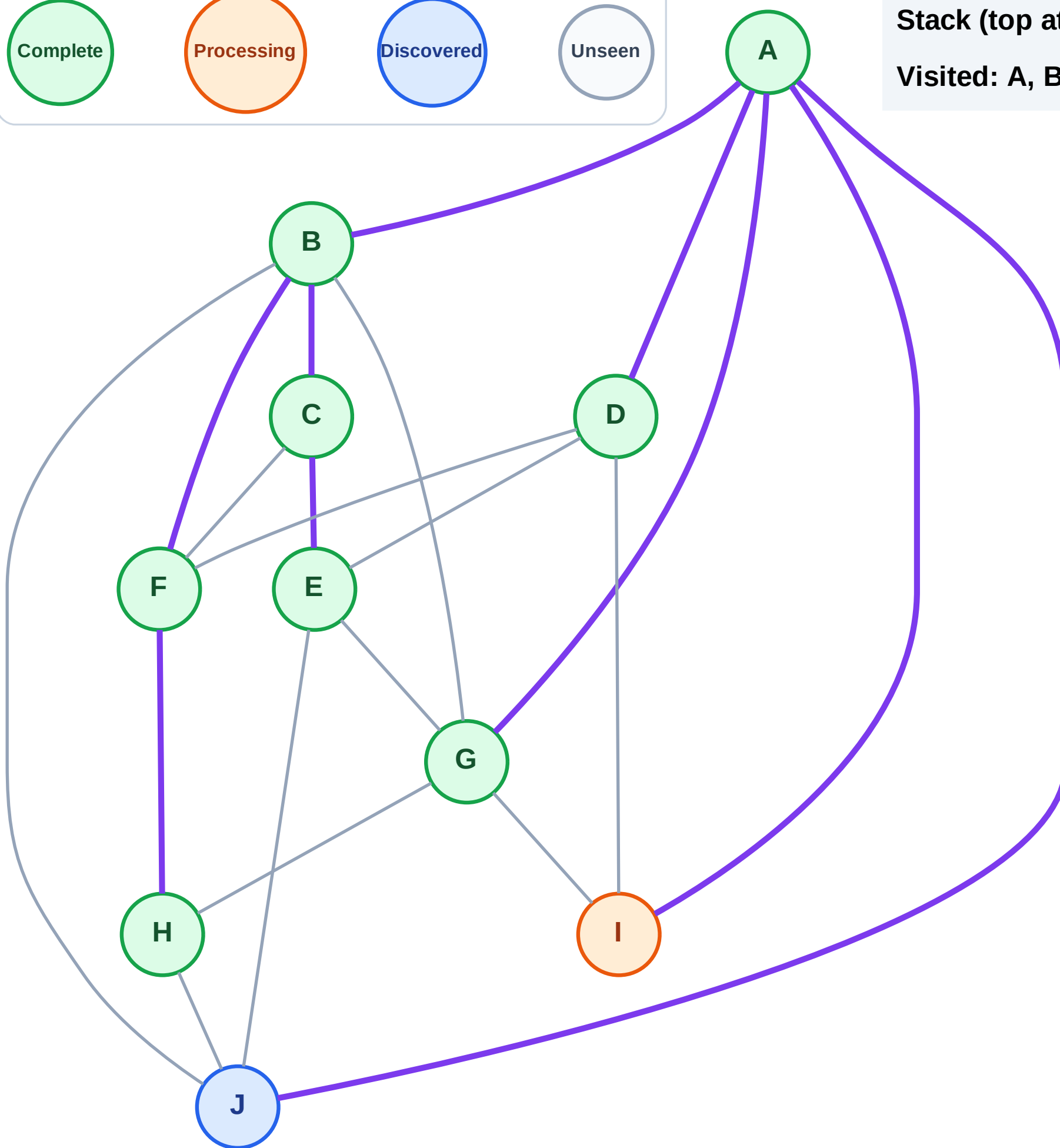
Processing

Discovered

Unseen

Stack (top at right): J

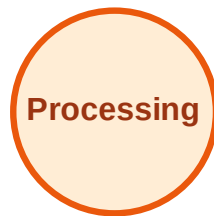
Visited: A, B, C, D, E, F, G, H



DFS Traversal

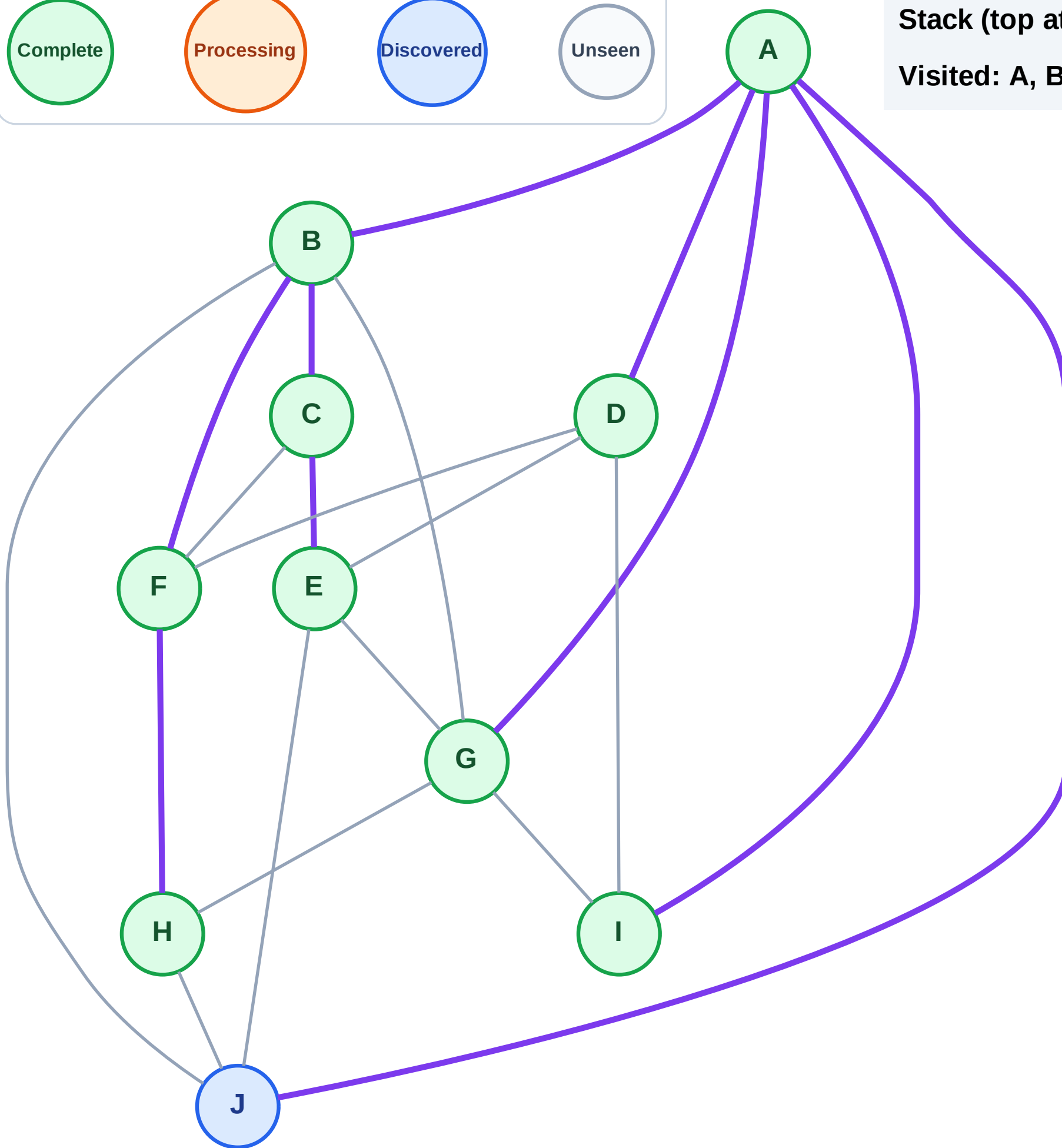
Step 19: No new neighbors from I

Legend



Stack (top at right): J

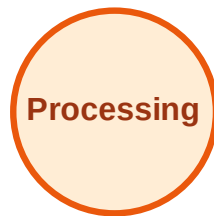
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

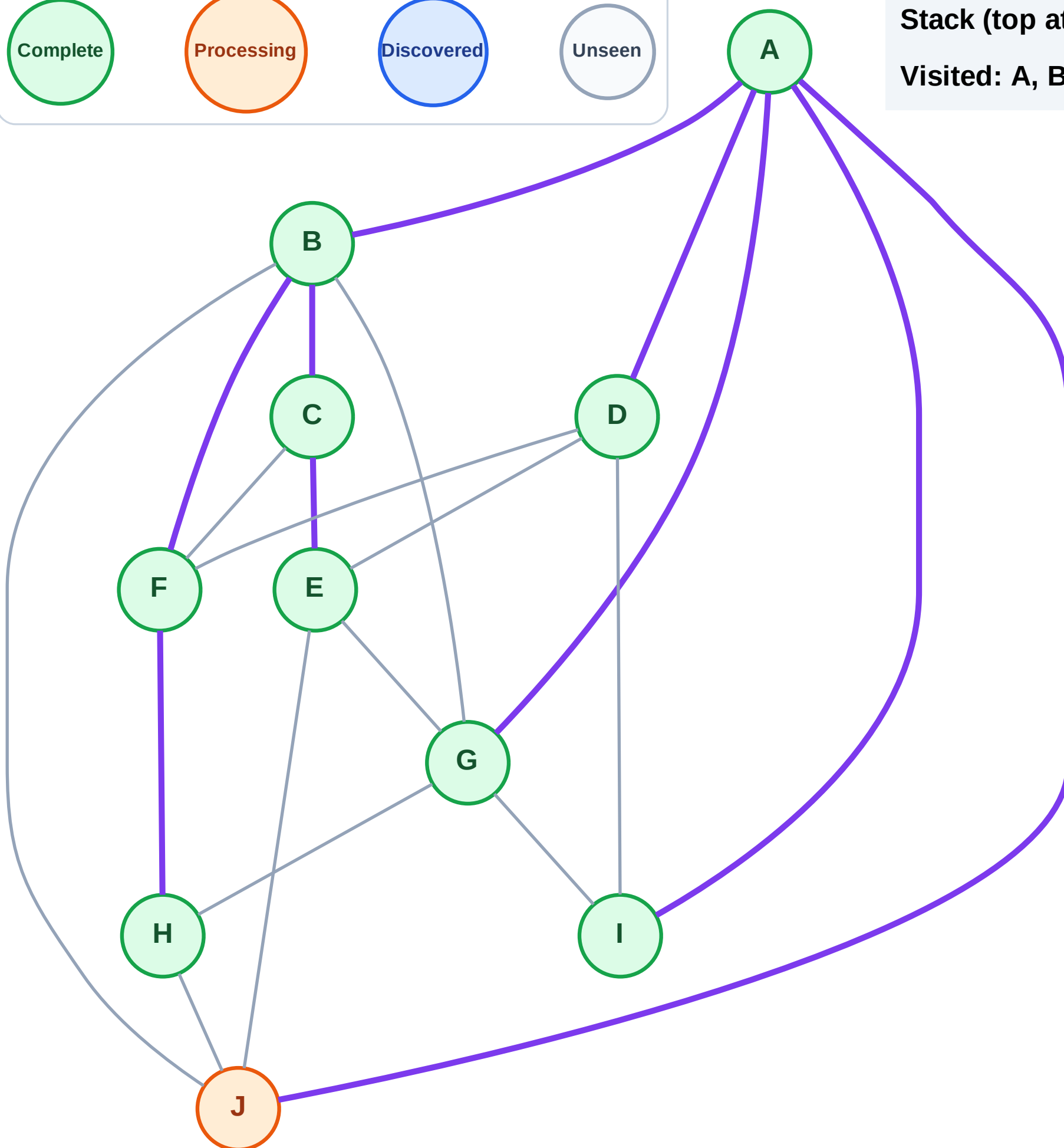
Step 20: Process node J

Legend



Stack (top at right): (empty)

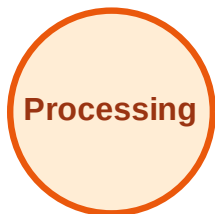
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

